



UNIVERSITY OF
LINCOLN

SCHOOL OF MATHEMATICS
AND PHYSICS

Mathematics of Ripples on Water



23rd April 2018

Scientific Report in the 3rd Year Module

MTH3005M Mathematics Project

Abstract

The aim of this project is to discuss the characteristics and motion of different systems of fluid and how limits of certain variables related to the fluid's motion can result in a substantial impact upon the shape and velocity of the waves that these fluids produce. Throughout this project there will also be a discussion on how external influences, such as an insoluble fluid on the surface, can affect the waves of the fluid. Finally this project will aim to produce visual representations of the fluids motion when possible by using mathematical software which will allow for a better understanding of topics discussed.

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1 Introduction

1.1 Overview of the Project

Over the course of this project, we will begin to develop a more in depth understanding of the fluids which dominate the surface of our planet. We will begin the project with a quick review of the fluid dynamics course covering the fundamental topics which the reader will be required to have an understanding of to fully engage with the topics discussed later in the project. This prerequisite understanding will not only provide the reader with the governing equations of motion for fluids but it will also attempt to give a full derivation of these equations. The reason we will take the time to derive these is due to the assumption that with the knowledge obtained from this analysis the reader will have a better understanding of the general topic which will result in more involved material for the reader overall. The prerequisite understanding section should also give the reader an indication of the type of analysis which will feature heavily throughout the project and help the reader to familiarise themselves with how the project will tackle derivation of equations throughout with derivations which are to a lower level of understanding.

Once we have completed the section on prerequisite understanding we will move on to our analysis of the motion of a fluid in gravity waves. During this section, we will provide a full derivation of the paths of fluid particle and velocity of wave propagation in both deep and shallow water. The project will be using the equations of motion derived previously and applying these to sets of initial and boundary conditions to obtain the desired equations. Throughout the analysis of both deep and shallow water, we will provide a visual representation of the paths that the fluid particle will take, this will provide the reader with a better understanding of how the fluid will move in different systems. The project will also show the reader how the fluid particles motion can change with differing positions in both the x and z axes. During the discussion of wave propagation, there will be a small real world application which will provide an explanation to what we see on shores after a storm out at sea.

Once we have derived the equations of motion for these gravity waves we will move our focus on to fluids which possess notable values of viscosity, fluids such as glycerine for example. This analysis will show the reader how a change in the fluid in which the waves are generated in can yield a substantial change in the overall motion of the waves. This section, as before, will provide the reader with visual representation of this damping effect in fluids which will have a small and large dynamic viscosity, η . This visual representation is significant because the equations alone do not show the extent that the viscosity influences the damping.

Once the reader has a firm understanding of the gravity waves and damping the project will begin the analysis of capillary waves or ripples. During this final section of mathematical analysis, we will develop our understanding of the smallest and most beautiful waves that we can find in nature and how these waves can influence the larger waves which we have previously derived. We will begin with a discussion on the Laplace equation which will develop the readers understanding of how we can derive equations that

will describe surface tension, the fundamental force for capillary waves which defined their unique shape and hydrodynamic properties. During our derivation of the Laplace equation, we will use some simple geometry and well-known formulae found in physics to find a relation between radii of curvature and displacement from the plane. Using the knowledge that the reader has developed from the Laplace equation the project will continue to derive the equations of motion governing capillary waves in a plane. The aim of this section is to show the reader how some basic characteristics can affect waves motion. Finally, this section will discuss how these special cases of waves are damped differently to that of a gravity wave using analysis techniques that the reader has developed previously when discussing the damping of gravity waves. This will lead to a discussion of why the damping coefficient is different for these types of waves.

Finally, in the last section of the project, the reader will begin to develop an understanding of how imperfections in the conditions surrounding the fluid can cause a difference in the characteristics of the wave. During the course of this section, the reader should expect less derivation of equations and more of a general discussion of how these external influences can affect the waves. The reason for this change is due to the difficulty in deriving equations for random events and we believe that the reader's time will be better spent developing an understanding of why the wave characteristics differ instead of a mathematical derivation of how. The discussion will allow the reader to have a better understanding of the motion of waves in real world applications due to the real-world never having perfect conditions unless the system is produced in a laboratory. The reader will also begin to develop an understanding of what we as humans have been doing for centuries to manipulate the fluid for our convenience.

1.2 Literature Review

We will here provide the reader with a review of the literature that we have used throughout the project. Whilst studying the resources available to us throughout the course of the year we have found that it is difficult to find a resource that is focused entirely on the area of mathematics we will be discussing in this project. We have found that it is either the case that the content has been printed in an academic resource, such as "Fluid Mechanics" by Landau and Lifshitz, or we find similar work in journal articles. In the case of the academic resources, we found that in general they only provided the derivation of some of the key results but lacked a discussion of said results. This may be due to the resource having to provide a large variety of topics which falls under the title of fluid dynamics and cannot be specialised. In the case of the journal articles, we tend to find that the paper aims to focus on one specific area inside what would be seen on a page of the academic resource. This generally resulted in the target audience of the journals to be to a higher level of mathematics than this project is being written for.

The difference between the aims of both types of resources may also be due to how technology has progressed over the years. We tend to find the academic resources are considerably older than the journals this trend, we assume, stems from the increasing ability to visual systems on a computer. This is evident if you observe the number of figures in a journal article and an academic resource.

After reading multiple resources we found that this project finds itself nicely in between the content for academic resources and journals. The justification for this claim is due to this project specialising its focus on a slightly larger topic than a journal would but provided substantially more information on the topics than the academic resources do. We feel that this project can be used a bridge to connect the complexity of the journals with the accessibility of academic resources.

1.3 Notation

The notation I will be using in this report is as follows:

ρ Density
 p Pressure
 η Dynamic Viscosity
 ν Kinematic Viscosity
 v Velocity
 V Volume
 a Amplitude
 ϕ, φ Velocity Potential
 g Gravity
 t Time
 T Time Period
 λ Wavelength
 k Wave Number
 ω Angular Frequency
 γ Damping Coefficient
 α Surface-Tension Coefficient

2 Prerequisite understanding

We begin the project with the derivation of the governing equations of motion. These equations are crucial in the analysis of more complex fluid motion that we will be studying throughout this project and evidently is important that we take the time to derive said equations.

2.1 Equation of continuity

We begin our study of fundamental equations with the equation of continuity. This equation mathematically represents the condition that mass must be conserved in fluid motion, it is a simple equation but one that is very important in any study of fluids.

If we consider a small volume, dV , which is enclosed by a surface, S , we know from a well known simple equation that the total mass of the fluid is given by $\int_V \rho dV$. We can also find the value of total flow out of a volume, V , by evaluating the total flow through the surface S as these two statements are equivalent. We know that the total flow through

a surface is given by

$$\oint_S \rho \mathbf{v} \cdot d\mathbf{f}.$$

We can also find the decrease in time of the mass of the fluid in V to be

$$-\frac{\partial}{\partial t} \int \rho dV.$$

By equating the two equations we find

$$\frac{\partial}{\partial t} \int \rho dV = - \oint_S \rho \mathbf{v} \cdot d\mathbf{f}. \quad (1)$$

Using Green's formula which is defined as

$$\oint_S \mathbf{a} \cdot d\mathbf{f} = \int_V \nabla \cdot \mathbf{a} dV, \quad (2)$$

and applying this to (1) we arrive at

$$\int_V dV \left[\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \mathbf{v} \right] = 0. \quad (3)$$

As the volume is none zero we can simplify equation to the following

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \mathbf{v} = 0, \quad (4)$$

which is the equation of continuity for a fluid. It is important to note this is a general equation of continuity. During the course of this project, we will see two more instances where we encounter an equation of a different form that is still named the equation of continuity. This reoccurring naming of equations is not a mistake from the project the equation can simply take different forms for different systems, however, the function of the equation remains the same.

2.2 Euler's equation

The next equation of motion we will derive is Euler's equation. Originally being published by Swiss mathematician Leonhard Euler in his article named "Principes généraux du mouvement des fluides", the equation provides us with a relation between velocity, density and pressure for ideal fluids which express steady flow.

We begin by considering a volume, V , of a fluid. We can express the total force acting on the volume to be

$$- \oint_S p d\mathbf{f}.$$

From the definition of force we have $\mathbf{F} = \oint_S d\mathbf{F}$ in our liquid. We also use the definition of pressure, $d\mathbf{F} = -pd\mathbf{f}$, which provides us with

$$\mathbf{F} = \oint_S d\mathbf{F} = - \oint_p d\mathbf{f}. \quad (5)$$

If we now apply Green's formula to the right hand side of equation (2) we find

$$\mathbf{F} = \oint_S d\mathbf{F} = - \int_V \nabla p dV. \quad (6)$$

If we consider an element of mass, m , we have

$$m\mathbf{a} = \int_V \mathbf{a} dm = \int_V \rho \frac{d\mathbf{v}}{dt} dV. \quad (7)$$

If we now use, the well known, Newtons second law, $\mathbf{F} = m\mathbf{a}$ and substitute equations (6) and (7) into the formula we arrive at

$$\int_V \rho \frac{d\mathbf{v}}{dt} dV = - \int_V \nabla p dV,$$

which can be simplified to the following

$$\rho \frac{d\mathbf{v}}{dt} = -\nabla p. \quad (8)$$

We know that $\frac{d\mathbf{v}}{dt}$ is the rate of velocity for a moving fluid particle and $d\mathbf{v}$ is an exact differential so by definition we have that

$$\frac{d\mathbf{v}}{dt} = \frac{\partial \mathbf{v}}{\partial t} + \frac{dx}{dt} \frac{\partial \mathbf{v}}{\partial x} + \frac{dy}{dt} \frac{\partial \mathbf{v}}{\partial y} + \frac{dz}{dt} \frac{\partial \mathbf{v}}{\partial z}.$$

We also know from the definition of velocity that $\mathbf{v} = (\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt})$ and from the definition of gradient we have $\nabla = \frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}$. If we apply these two definitions to our equation for acceleration we obtain

$$\frac{d\mathbf{v}}{dt} = \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v},$$

applying this to equation (8) we arrive at

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = \frac{\nabla p}{\rho} \quad (9)$$

this is Euler's equation. However this is without the presence of gravity if we wanted it with gravity we simply need to add the term $\mathbf{g}$ which is Euler's equation in the presence of gravity which can be seen below

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = \frac{\nabla p}{\rho} + \mathbf{g}. \quad (10)$$

2.3 Bernoulli's equation

The final equation of motion that we will cover in this section is Bernoulli's equation. Bernoulli's principle was originally published in Daniel Bernoulli's book "Hydrodynamica", however, it was Euler who wrote the equation into the form that we know today as Bernoulli's equation. This equation tells us how the speed of the fluid relates to the pressure of the fluid with density and gravity also considered.

We begin our derivation of Bernoulli's equation by using the definition of steady flow which states $\frac{\partial \mathbf{v}}{\partial t} = 0$. This property is a result of the velocity not changing in time it can, however, change in space so for this system we have the following $\mathbf{v}(x, y, z, t) = \mathbf{v}(x, y, z)$.

We know from section 2.1 that Euler's equation for a fluid is defined as

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = \nabla p + \mathbf{g}$$

If we consider an incompressible fluid, $\rho = \text{constant}$, with a system that has an absence of gravitational field, $\mathbf{g} = 0$, and apply the definition $(\mathbf{v} \cdot \nabla) \mathbf{v} = \frac{1}{2} \nabla v^2 - \mathbf{v} \times (\nabla \times \mathbf{v})$, proof of which can be found in the appendices, to Euler's equation we obtain

$$\frac{1}{2} \nabla v^2 - \mathbf{v} \times \nabla \times \mathbf{v} = -\nabla \left(\frac{p}{\rho} \right).$$

We now introduce the idea of a streamline. A streamline is a line whose tangent gives the direction of the velocity at that moment. If we consider the above equation along a streamline and introduce a tangent unit vector $\hat{l}$. From the definition of a tangent we see that $\hat{l}$ is parallel to $\mathbf{v}$ on a streamline, hence the element $(\mathbf{v} \times \nabla \times \mathbf{v})$ is perpendicular to $\hat{l}$. Therefore if we multiply the equation through by the dot product of $\hat{l}$ we find

$$\left(\nabla \left(\frac{1}{2} v^2 + \frac{p}{\rho} \right) \right) \cdot \hat{l} = 0$$

on a streamline. We also have that the unit vector $\hat{l}$ by definition is none zero hence we can simplify to

$$\nabla \left(\frac{1}{2} v^2 + \frac{p}{\rho} \right) = 0.$$

As the gradient of $\frac{1}{2}v^2 + \frac{p}{\rho}$ is equal to zero we can then deduce that

$$\frac{1}{2}v^2 + \frac{p}{\rho} = C \quad (11)$$

where C is a constant. This is Bernoulli's equation for an incompressible fluid without the presence of gravity. In the case of gravity being present using similar procedures we get that Bernoulli's equation in presence of gravity is

$$\frac{1}{2}v^2 + \frac{p}{\rho} + gz = C \quad (12)$$

where C again is a constant.

2.4 Potential flow

Our final section in the prerequisite understanding chapter is on potential flow. Potential flow is not an equation that requires deriving it merely a definition of a fluid when certain conditions are satisfied.

For a flow to be considered potential it must satisfy two conditions: the fluid must be incompressible, $\nabla \cdot \mathbf{v} = 0$, and must have zero vorticity, $\omega = \nabla \times \mathbf{v} = 0$. From these definitions, it follows that the velocity for a potential flow can be written as $\mathbf{v} = \nabla \varphi$ as we know that $\nabla \times (\nabla \varphi) = 0$. We can also take the divergence of $\mathbf{v}$ which gives us $\Delta \varphi = 0$.

By taking Euler's equation (9) and make the substitution $(\mathbf{v} \cdot \nabla) \mathbf{v} = \frac{1}{2} \nabla v^2 - \mathbf{v} \times (\nabla \times \mathbf{v})$ as we did before in the section on Bernoulli's equation, we arrive at the following

$$\frac{\partial \mathbf{v}}{\partial t} + \frac{1}{2} \nabla v^2 - \mathbf{v} \times (\nabla \times \mathbf{v}) = -\frac{1}{\rho} \nabla p.$$

Using the definition of $\mathbf{v} = \nabla \varphi$ for potential flow and substituting this into the equation we obtain

$$\nabla \left(\frac{\partial \varphi}{\partial t} + \frac{1}{2} v^2 + \frac{p}{\rho} \right) = 0,$$

which finally simplifies to

$$\frac{\partial \varphi}{\partial t} + \frac{1}{2} v^2 + \frac{p}{\rho} = f(t), \quad (13)$$

where $f(t)$ is a function of time.

This concludes our section on the prerequisite understanding. We hope that content discussed in this section of the report will help the reader to now continue with the main body of text.

3 Gravity Waves

3.1 Introduction to Gravity Waves

Gravity waves arise when some disturbance in the surface of the water causes a disruption to the equilibrium of the water. In positions in which the water rises above the equilibrium, gravity will act upon the water and pull it down as the density of water is more than it is of the atmosphere. The inertia that is acquired from this falling motion reduces the water to below the equilibrium hence causing the water to again bob up above the equilibrium. This process continues until the fluid returns to equilibrium and displays an oscillating motion in the z -axis. The most obvious visualisation of this process is the waves found in the sea. We know that waves in the sea do not possess motion that is solely in the z -direction, however, they also propagate horizontally. To understand why this horizontal motion occurs, we consider an object that is dropped into the water that is at equilibrium, when the object enters the water the pressure surrounding the object is higher than that of the surrounding fluid and because of this high relative pressure water accelerates away from the object in all directions causing a propagation of fluid. It is important to note that these waves are still affecting the interior of the fluid but at a lower amount the deeper that the fluid is.

3.2 Linearisation

We begin our analysis of gravity waves by considering how we can use the mathematical process of linearisation to allow the method of deriving equations of motion to be much simpler for us. Throughout the duration of this section on gravity waves will consider waves that have a very small velocity of moving fluid particles. Mathematically this condition implies that $(\mathbf{v} \cdot \nabla) \cdot \mathbf{v}$ is at least of two orders smaller than $\frac{\partial \mathbf{v}}{\partial t}$. From this condition, we can consider the term $(\mathbf{v} \cdot \nabla) \cdot \mathbf{v}$ to be negligible and therefore we can omit this element from the equations in this subsection.

Below we have a simple solution of the sine function plotted in 3D.

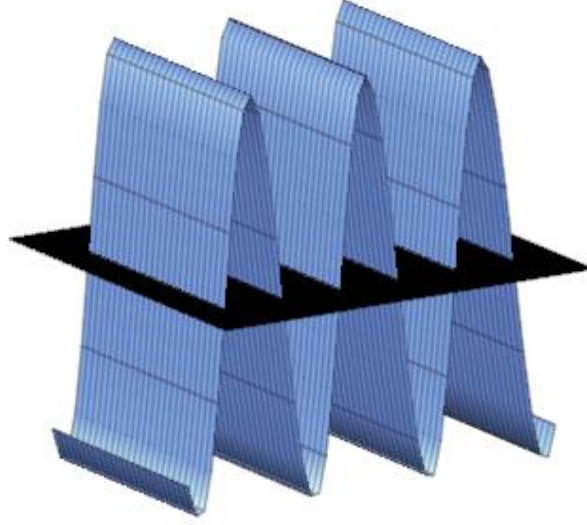


Figure 1 - A 3D plot of a Sinusoidal Wave for a Varying x Variable. The Black Plane Represents the Fluid at Equilibrium.

This plot shows what a basic system which displays wave motion would look like, this is what we assume the wave motion would look like without any mathematical manipulation.

We continue by declaring some well know variables in physics: amplitude, a , wavelength, λ , and the period of the oscillations of fluid particles in the wave, τ . If we consider a fluid particle resting at the peak of the wave in figure 1, it is clear that during a period of a single oscillation the particle can at most travel a distance of $4a$ and a minimum of 0. From the definition of an order it is evident that a is the same order as $4a$ hence we have that during a time interval a fluid particle travels a distance that is of the order a . From this we can determine that the velocity of the fluid particle is of order $\frac{a}{\tau}$. The velocity of the wave varies during time intervals of the order T and distances of the order λ hence the time derivative of the velocity is of order $\frac{v}{\tau}$ and the spacial derivatives are of order $\frac{v}{\lambda}$. Applying this to the original condition we stated,

$$(\mathbf{v} \cdot \nabla)\mathbf{v} \ll \frac{\partial \mathbf{v}}{\partial t},$$

we find the following

$$\frac{1}{\lambda} \left(\frac{a}{\tau} \right)^2 \ll \frac{a}{\tau} \cdot \frac{1}{\tau}$$

which can be simplified to,

$$a \ll \lambda.$$

This equation signifies that for a wave with a small velocity of moving fluid particles we must have that the amplitude is significantly smaller than the wavelength. We assume that the fluid that we are considering is incompressible, then it follows that we have a potential flow due to the term $(\mathbf{v} \cdot \nabla)\mathbf{v}$ being neglected.

We now consider the alternative form of Euler's equation (13), it is clear that we can omit the $\frac{1}{2}v^2$ term without loss of general accuracy in our analysis as the velocity component is defined as being small which would then be halved and be squared leaving the velocity incredibly small hence can be disregarded. Finally due to the presence of a gravity field, we also add the term gz leaving us with the equation of motion for our fluid given by

$$\frac{\partial \varphi}{\partial t} + \frac{p}{\rho} + gz = f(t), \quad (14)$$

where $f(t)$ is a constant in space. Let us now define a new notation, ζ , for a position on the z -axis of a fluid particle. We define $\zeta = 0$ when our particle is on the surface of the fluid at equilibrium. Setting this notation allows us to define any displacement of a particle from equilibrium equal to ζ . Clearly, ζ is a function of x, y and t . We define p_0 be a constant pressure that is applied on the surface of the fluid, applying this definition to equation (14) we find that at the surface we have

$$g\zeta + \frac{\partial \varphi}{\partial t} + \frac{p_0}{\rho} = f(t).$$

Clearly, for this fluid, we are left with two constants in our equation of motion. We wish to simplify our equation of motion by removing these constants, to do this we simply redefine our potential

$$\varphi \rightarrow \varphi - \left(\frac{p_0}{\rho}\right)t$$

which we can do without changing the velocity as $\mathbf{v} = \nabla \varphi$ and as the added term is a constant with respect to x, y, z applying the gradient function to this will remove the constant. Redefining the potential gives us

$$g\zeta + \frac{\partial \varphi - (\frac{p_0}{\rho})t}{\partial t} + \frac{p_0}{\rho} = f(t).$$

Finally, we will define $f(t) = 0$, which again we can do without loss of general accuracy, applying this we can see that our equation simplifies to

$$\zeta = -\frac{1}{g} \left(\frac{\partial \varphi}{\partial t} \right)_{z=\zeta}. \quad (15)$$

By definition, we know that

$$v_z = \frac{\partial \zeta}{\partial t} = \left(\frac{\partial \varphi}{\partial z} \right)_{z=\zeta},$$

so by taking the first time derivative of (15), we obtain an equivalence relation given by

$$-\frac{1}{g} \left(\frac{\partial^2 \varphi}{\partial t^2} \right)_{z=\zeta} = \left(\frac{\partial \varphi}{\partial z} \right)_{z=\zeta}$$

or,

$$\left(\frac{1}{g} \frac{\partial^2 \varphi}{\partial t^2} + \frac{\partial \varphi}{\partial z} \right)_{z=\zeta} = 0.$$

As we have defined the oscillations to small for this case we can take the derivatives for $z = 0$ rather than $z = \zeta$. So the system of equations we are required to solve to obtain the motion are

$$\left(\frac{1}{g} \frac{\partial^2 \varphi}{\partial t^2} + \frac{\partial \varphi}{\partial z} \right)_{z=0} = 0 \quad (16)$$

and

$$\Delta \varphi = 0. \quad (17)$$

3.3 Deep Water Solution

There are three different types of gravity waves which we will encounter in nature; deep water, shallow water and intermediate gravity waves. This project will consider the deep water and shallow water limits in this section. We will see that these limits will impact the shape of the fluid particles path and the velocity of wave propagation. We will begin with the deep water solution.

3.3.1 Path of Fluid Particle

The deep water limit states that $\lambda \ll H$ where λ is the wavelength and H is the depth of the water. At this limit, we can neglect the boundary conditions at the bottom and the sides of the fluid as the external waves produced will have decayed before it reaches the bottom surface. For our wave, we are making the assumption that we have the fluid propagating along the x -axis and is uniform in the y -direction as figure one shows, with these assumptions we find that our solution will be independent of the y -component. It is reasonable to assume that our solution for velocity potential will be in the form of a simple period function of time and the x coordinate hence we have the following equation for the potential

$$\phi = f(z) \cos(kx - \omega t)$$

where ω is the angular frequency of the wave and k is the wave number which is equivalent to $\frac{2\pi}{\lambda}$. Recalling the definition $\Delta \varphi = 0$ and substituting our value for the potential into said definition we find

$$\frac{\partial^2 f(z) \cos(kx - \omega t)}{\partial z^2} - k^2 f(z) \cos(kx - \omega t) = 0.$$

Clearly, we begin our analysis by finding a solution for $f(z)$. To find a solution for this function we fix a value of time such that $\cos(kx - \omega t) \neq 0$ which allows us to divide

the equation through by $\cos(kx - \omega t)$. Doing this leaves us with the following relation for $f(z)$

$$\frac{\partial^2 f(z)}{\partial z^2} - k^2 f(z) = 0.$$

The solution to this relation is given by

$$f(z) = c_1 e^{kz} + c_2 e^{-kz}.$$

From the deep water limit, we know that at the bottom surface of the fluid we would observe no movement of fluid. This limit allows us to define the boundary condition at the bottom surface given by $z \rightarrow -\infty$ $f(z) = 0$. If we apply this boundary condition to our solution for $f(z)$ we find

$$c_1 e^{kz}|_{z \rightarrow -\infty} + c_2 e^{-kz}|_{z \rightarrow -\infty} = 0$$

it is clear that the first term will decay away to 0 as $\rightarrow -\infty$ as it will become a power of a negative infinity, however, for the second term it is clear that e^{-kz} approaches infinity for the limit. Therefore the second term does not satisfy our boundary condition except for $c_2 = 0$ and we are left with the following solution for the velocity potential

$$\phi = c_1 e^{kz} \cos(kx - \omega t). \quad (18)$$

We can now apply our value for the velocity potential into (16) which yields

$$-\frac{1}{g} \omega^2 c_1 e^{kz} \cos(kx - \omega t) + c_1 k e^{kz} \cos(kx - \omega t) = 0$$

or by finding a value for time, t , such that $\cos(kx - \omega t) \neq 0$ we can divide through by this value to yield

$$\omega^2 = gk. \quad (19)$$

This relation is called the dispersion relation. The dispersion relation describes the relationship between the wave number and frequency in a fluid.

All that remains to be found in this section is the path of fluid particles. To find these equations we are first required to find a formula for v_x and v_z . From simple linear algebra we know that $v_x = \frac{\partial \phi}{\partial x}$ and $v_z = \frac{\partial \phi}{\partial z}$ so to find the motion in these directions we simply require to take the spatial derivatives of the potential. Doing such yields the following solutions

$$v_x = -A k e^{kz} \sin(kx - \omega t), \quad v_z = -A k e^{kz} \cos(kx - \omega t).$$

It is clear from these equations that the velocity in both the x and z directions decay exponentially as z becomes more negative, visually you will observe that as we consider fluid particles that are deeper into the fluid the velocity of the waves decrease rapidly. We now wish to determine the path of the fluid particles in the wave. To do this we begin

by denoting the coordinates of a moving fluid particle, which is not fixed in space, by x, z and x_0, z_0 , where the values of x_0 and z_0 represent the centre of the fluid motion. For the right-hand side of the equations we can approximate the values of the variables in the trigonometric functions to be $x = x_0$ and $z = z_0$ as the oscillations of the wave are small. We know by definition that $v_x = \frac{dx}{dt}$ and $v_z = \frac{dz}{dt}$ so we are only required to integrate the equations with respect to time to find the values for X and Z . Integrating yields

$$X = x - x_0 = -A \frac{k}{\omega} e^{kz_0} \cos(kx_0 - \omega t) \quad (20)$$

$$Z = z - z_0 = -A \frac{k}{\omega} e^{kz_0} \sin(kx_0 - \omega t). \quad (21)$$

We wish to find a visual representation of this fluid particle, to do this we can apply our values for X and Z to the equation of a circle, $a^2 + b^2 = r^2$. Making this substitution we arrive at the following

$$X^2 + Z^2 = \left(\frac{Ak}{\omega} \right)^2 e^{2kz_0}. \quad (22)$$

Which is a solution that is in a form that we can plot using Maple.

Below is a graph plotted in Maple to the graphical solution for (22), we have used the following values to plot this solution, $z_0 = \{0, -0.7, -1\}$, $x_0 = \{0, \frac{1}{2}\pi\}$, $\lambda = 4\pi$, $k = \frac{1}{2}$, $A = 1$, $\omega = 4.905$

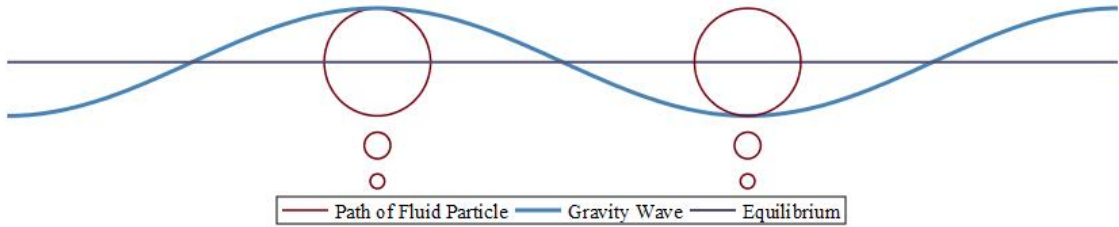


Figure 2 - Path of Fluid Particles in Deep Water Solution.

As we can see from the solution the path of each fluid particle is given by circles about the point (x_0, z_0) with a radius that becomes exponentially smaller with decreasing values of z_0 , i.e. the deeper the fluid particle is in the system. We also note that this solution does not depend on x_0 hence at any point along the x you will find that the solution of motion will be the same for a set value of z_0 .

3.3.2 Velocity of Wave Propagation

The final piece of information we will derive for deep water gravity is the velocity of the wave propagation. To find this we use the formula for the velocity of wave propagation which is given by $U = \frac{\partial \omega}{\partial k}$. If we substitute our value for ω into this formula we find that the velocity of wave propagation is given by

$$U = \frac{\partial \sqrt{kg}}{\partial k} = \frac{1}{2} \sqrt{\frac{g}{k}} = \frac{1}{2} \sqrt{\frac{g\lambda}{2\pi}}. \quad (23)$$

As we can see the only variable that the equation for U depends on is wavelength, λ . From this, it is evident that for an increasing wavelength we would observe an increasing velocity of wave propagation. Below you will see a plot to show the relationship between U and λ .

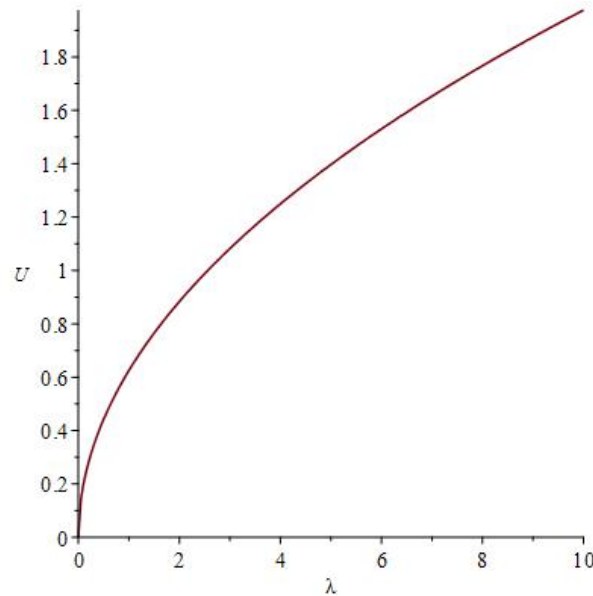


Figure 3 - Relation Between Velocity of Wave Propagation and Wavelength.

From this figure, we can see that initially a small increase in the wavelength of a wave will result in a significantly larger value for U , however, when we consider a large wavelength a small change in the wavelength will not provide as large of an impact on the velocity of wave propagation as it would for a relatively small wavelength. This effect plays an important role in nature and for humans in recent history where this effect would give an indication of a storm out at sea. When there is a large storm in the ocean it results in a production of large gravity waves, as we have found in this section when these waves are produced the waves that possess a larger wavelength propagate faster than waves with a smaller wavelength. The significance of this effect is that large gravity waves with a large wavelength will appear at the coastline much quicker than short wavelength waves, so when people observed large wavelength waves at a coastline it was an adequate assumption that there was a large storm close by. Clearly, this technique would not be applied today with the use of satellites to track conditions around the globe but it is still an interesting application of the work done in this subsection.

3.4 Shallow Water Solution

In the previous section we considered waves with an infinite depth and therefore a small wavelength in relation to the depth, in making these assumptions it was sufficient to neglect the boundary conditions which appeared at the bottom surface of the fluid as the waves will have decayed before it could reach said surface. In this section, however, we will be analyzing the behaviour of a fluid that has a depth that is roughly $d < \frac{\lambda}{20}$ where λ is wavelength. As we will be considering fluid with such properties we will, in

this case, have to consider the boundary conditions at the bottom surface. As before we will work towards gaining a visual representation of the path of the fluid particle and the velocity of wave propagation in some different instances of shallow water.

3.4.1 Path of a Fluid Particle in Shallow Water

We begin as before by defining that the fluid we are considering in this section is irrotational and therefore the fluid will pass through solid layers freely. As the fluid is irrotational we know that the fluid must satisfy the Laplace equation for two-dimensional flow,

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial z^2} = 0$$

We also make the assumption, in this case, that the base is solid, impermeable and horizontal so that the base does not add or remove energy from the fluid. However, it would be suitable to use these calculations for a slightly sloped base as an approximation. As before we make the assumptions that the pressure along the fluid air boundary is constant across the entirety of the surface, the fluid is homogeneous and incompressible and that the surface tension is negligible. We also assume that the amplitude is small in comparison to the wavelength as we did before in the deep water solution. With these conditions imposed we begin to find a model to describe the path of the fluid particles.

There are two boundary conditions along with the Laplace equation which we need to be able to define the potential in this instance and this is where we will begin our analysis. At the bottom surface of the system we know that the fluid will have no flow acting in the perpendicular direction hence we have that

$$v_z = \frac{\partial \varphi}{\partial z} = 0$$

for $z = -d$. This is the boundary condition at the bottom surface.

Bernoulli's equation for unsteady irrotational flow is given by

$$\frac{1}{2}(v_x^2 + v_z^2) + \frac{p}{\rho} + gz + \frac{\partial \varphi}{\partial t} = 0.$$

The pressure at the free surface, we know, is equal to zero, $p = 0$. We now define our boundary at the free surface to be given by

$$\frac{1}{2}(v_x^2 + v_z^2) + gz + \frac{\partial \varphi}{\partial t} = 0$$

for $z = \zeta$. If we now linearise the simplified Bernoulli's equation, we can see that the term $\frac{1}{2}(v_x^2 + v_z^2)$ is negligible and we can, therefore, omit the term from the boundary condition. Applying this simplified Bernoulli's equation to the equilibrium of the fluid

we have our second boundary condition given by

$$g\zeta + \frac{\partial\varphi}{\partial t} = 0 \quad (24)$$

for $z = 0$. This is called the dynamic boundary condition.

Using Laplace's formula, the boundary condition at the bottom surface and the dynamic boundary condition, we can find the velocity potential to be given by

$$\phi = \frac{gH}{2\omega} \frac{\cosh(k(d+z))}{\cosh(kd)} \sin(kx - \omega t). \quad (25)$$

To continue with this procedure we need to find the dispersion relation for shallow water waves as we did for deep water waves. To do this we first need to impose another boundary condition, the kinematic boundary condition. This condition relates the vertical component of the fluid velocity for a fluid particle at the surface to the position on the surface and is defined as

$$v_z = \frac{\partial\zeta}{\partial t} + v_x \frac{\partial\zeta}{\partial x}$$

for $z = \zeta$. As we did before, we want to apply this boundary condition to the equilibrium level of the fluid and linearise the expression, doing so yields

$$\frac{\partial\varphi}{\partial z} = v_z = \frac{\partial\zeta}{\partial t} \quad (26)$$

for $z = 0$.

By taking the first time derivative of the dynamic boundary condition (24) and making the substitution $\frac{\partial\zeta}{\partial t} = \frac{\partial\varphi}{\partial z}$ we find the following

$$g \frac{\partial\varphi}{\partial z} + \frac{\partial^2\varphi}{\partial t^2} = 0$$

for $z = 0$. Substituting in our value for the velocity potential, (20), we find

$$g \frac{\partial \frac{gH}{2\omega} \frac{\cosh(k(d+z))}{\cosh(kd)} \sin(kx - \omega t)}{\partial z} + \frac{\partial^2 \frac{gH}{2\omega} \frac{\cosh(k(d+z))}{\cosh(kd)} \sin(kx - \omega t)}{\partial t^2} = 0.$$

Once the differential has been performed we first need to consider a time, t , such that $\cosh(k(d+z))\sin(kx - \omega t)/\cosh(kd) \neq 0$ so that we can divide through by this value to find the dispersion relation for this system. Solving for ω yields the following

$$\omega^2 = gk \tanh(k(d+z)).$$

As we have specified that $z = 0$ for the boundary condition we are left with

$$\omega^2 = gk \tanh(kd).$$

This is a more general dispersion relation to the one that we found earlier in our discussion of the deep water limit. It is easy to prove that the two values we have found for the dispersion relation are consistent at the deep water limit $d = \infty$, the value of $\tanh(kd) = 1$ hence the dispersion relation for deep water is given by $\omega^2 = kd$ which is consistent with what we found earlier. We also know from the shallow water limit, $kd \ll 1$, and hence our dispersion equation at the shallow water becomes

$$\omega^2 = gk^2d$$

as $\tanh(kd) = kd$ at this limit.

We have now obtained all the components we need to able to find an equation which describes the path of a fluid particle. We begin our final analysis in this section by restating a simple bit of notation for clarity.

$$v_x = \frac{\partial \phi}{\partial x} \quad v_z = \frac{\partial \phi}{\partial z}.$$

By making the substitution for the value of the velocity potential into this we obtain

$$v_x = \frac{gkH}{2\omega} \left[\frac{\cosh(k(d+z))}{\sinh(kd)} \right] \cos(kx - \omega t) \quad (27)$$

and

$$v_z = \frac{gkH}{2\omega} \left[\frac{\sinh(k(d+z))}{\sinh(kd)} \right] \sin(kx - \omega t). \quad (28)$$

This gives us a solution for the velocity components in both the x and z -directions and as we can see both components are comprised of 3 parts, the velocity at the surface of a fluid particle given by $\frac{gkH}{2\omega}$, the part in the square brackets which accounts for the depth of the water and the trigonometric function which determines the motion of the fluid particle as both the time, t , and position, x , changes.

There are two important cases we consider for the terms in the square brackets of (27) and (28) these being the shallow and deep water limits. We have already discussed the deep water limit in our previous section but to show the consistency of this method and the one previously used we shall quickly show the deep water limit. When considering deep water we know that $kd \gg 1$ so it follows that the values of the terms in square brackets are given by

$$\frac{\cosh(k(d+z))}{\sinh(kd)} = \frac{\sinh(k(d+z))}{\sinh(kd)} = e^{kz},$$

which by a simple substitution, we can see that this result is consistent with the one found in (18).

If we now consider the shallow water limit, $kd \ll 1$, using what we know about the trigonometric functions and small values it is easily shown that

$$\frac{\cosh(k(d+z))}{\sinh(kd)} = \frac{1}{kd}$$

and

$$\frac{\sinh(k(d+z))}{\sinh(kd)} = 1 + \frac{z}{d}.$$

Finally, by substituting this into our values for v_x and v_z we have

$$v_x = \frac{gH}{2\omega d} \cos(kx - \omega t) \quad (29)$$

and

$$v_z = \frac{gkH}{2\omega} \left(1 + \frac{z}{d}\right) \sin(kx - \omega t). \quad (30)$$

Finally, all that remains is to integrate the values of (29) and (30) with respect to t to obtain a value for X and Z as we did before. As we did in the deep water solution we will denote the coordinates of a moving fluid particle that is not fixed in space by x , z and x_0 , z_0 where again the values of x_0 and z_0 represent the centre of the fluid motion. We approximate the variables on the right hand side to be $x = x_0$ and $z = z_0$ as the oscillations of the wave are small. Integrating the equations yields

$$X = x - x_0 = -\frac{gH}{2d\omega^2} \sin(kx_0 - \omega t) \quad (31)$$

and

$$Z = z - z_0 = \frac{kgH}{2\omega^2} \left(1 + \frac{z_0}{d}\right) \cos(kx_0 - \omega t) \quad (32)$$

Again we will substitute the values for X and Z into the equation for a circle to obtain an equation which we can plot to find the motion of the fluid particle. Making this substitution yields

$$(d(X - x_0))^2 + \left(\frac{(Z - z_0)}{k(1 + z_0/d)}\right)^2 = \left(\frac{gH}{2\omega^2}\right)^2$$

Which is the form that we desire for us to be able to plot this motion of a fluid particle. By using Maple we can again find a visual solution which you can see below.

This figure shows the path of a fluid particle in shallow water and, for simplicity, we have defined the variables as follows: $d = 1$, $k = 0.01$, $H = 0.1$, $g = 9.81$, $\omega = \sqrt{9.81 * 0.01^2}$, $z_0 = \{0, 0.3, 0.55, 0.75, 0.9\}$ and $x_0 = \{0, 10\pi\}$

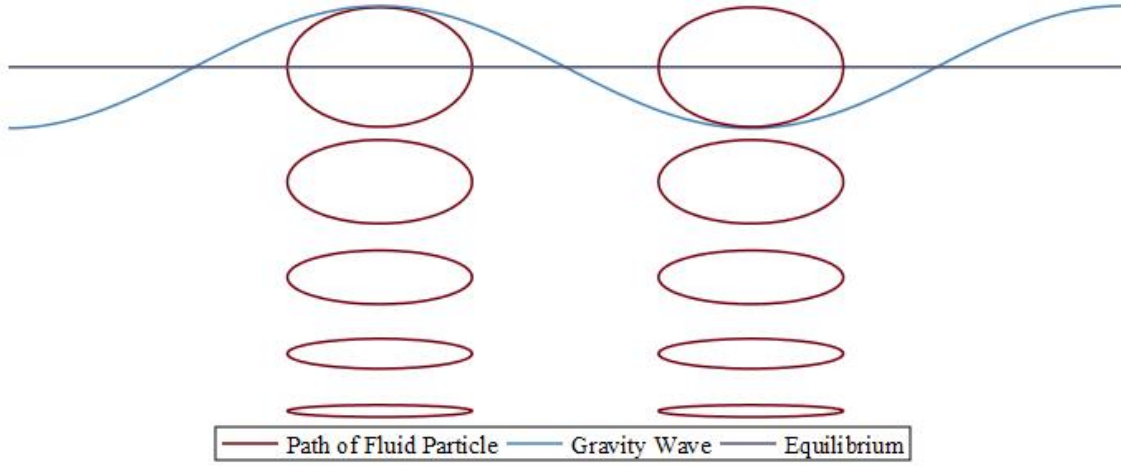


Figure 4 - Path of Fluid Particles in Shallow Water Solution.

As we can see from the figure above, the path of a fluid particle in shallow water has the same property as the deep water solution in terms of a change in the value of x_0 will result in no change in the shape of the path of the fluid particle. When we change the value of z_0 , however, we can see that the horizontal component of the fluid particle's path is unaffected but the vertical component of the fluid particle's path becomes smaller the more negative the value of z_0 is. We can see from our equations that at the base of the system, $z_0 = d$, that we have an undefined value as we would obtain $(\frac{Z-z_0}{0})^2$, this problem is one of the reasons in which we defined the boundary condition at the bottom surface. This boundary condition states that at the bottom surface we will have that the position of z will be a constant. From this result, it follows that the position of x will also be a constant hence at the bottom boundary we will have no movement as we did in the deep water solution.

3.4.2 Shallow Water Solution in a Channel

Now we have evaluated the motion of the fluid particle all that remains is to find the velocity of wave propagation and for this limit, we will consider two different systems of flow in shallow water. We will begin our analysis by finding the solution to the velocity of wave propagation in a channel.

To begin this analysis we assume that we have a system of fluid which resembles flow along an infinitely long channel where the fluid is moving along the x -axis. We shall here denote the cross-sectional area of the channel to be $S = S(x, t)$. In this subsection, we will consider longitudinal waves moving down the channel. For this type of motion, we know that the velocity component v_x is much larger than that of the velocity components v_y and v_z , hence for this system, we shall omit the velocity components v_y and v_z in the equations we shall be using. Now that we have described our system we continue by taking Euler's equation of motion (5) for the x -component of the flow

$$\frac{\partial v_x}{\partial t} + (v_x \cdot \frac{\partial}{\partial x})v_x = -\frac{1}{\rho} \frac{\partial p}{\partial x}.$$

By using similar procedures as the ones used in section **3.2.1** we can omit the term $(v_x \cdot \frac{\partial}{\partial x})v_x$, doing so leaves us with the following equation of motion

$$\frac{\partial v_x}{\partial t} = -\frac{1}{\rho} \frac{\partial p}{\partial x}. \quad (33)$$

We can also find the z component of the flow by again taking Euler's equation and omitting $(v_z \cdot \frac{\partial}{\partial z})v_z$ and $\frac{\partial v_z}{\partial t}$, as we consider v_z to be a small value, which yields

$$\frac{1}{\rho} \frac{\partial p}{\partial z} = -g. \quad (34)$$

By taking an integral with respect to z of both sides of (34) we arrive at

$$p = -\rho g z + C$$

where C is a constant. We also know that the pressure at the surface, $z = \zeta$, is given by $p = p_0$. Applying this condition to the integral we obtain

$$p = p_0 + \rho g (\zeta - z). \quad (35)$$

By substituting (35) into (33) we arrive at

$$\frac{\partial v_x}{\partial t} = -g \frac{\partial \zeta}{\partial x}, \quad (36)$$

as ζ is a function of time. To solve (36) and find the two unknown variables we consider a volume of fluid which is bounded between two plane cross sections of the channel which have a distance of dx apart from one another. In a unit of time, we know that a volume of $(Sv_x)_x$ will pass through one plane and a volume of $(Sv_x)_{x+dx}$ passes through the other. We know the change in volume between the two planes to be given by

$$\frac{\partial v_x}{\partial x} dx = (Sv_x)_{x+dx} - (Sv_x)_x.$$

However, as the fluid is incompressible we know that any change in volume must be due to a change in the level of the fluid so we can denote the change in volume per unit of time, $\frac{\partial S}{\partial t} dx$, to be

$$\frac{\partial S}{\partial t} dx = -\frac{\partial (Sv_x)}{\partial x} dx$$

or

$$\frac{\partial S}{\partial t} + \frac{\partial (Sv_x)}{\partial x} = 0. \quad (37)$$

This is the equation of continuity for this system.

By defining S_0 to be the cross-sectional area of the fluid at equilibrium and S' to be the change in the cross-sectional area, we have the following definition for cross-sectional

area $S = S_0 + S'$. Since, for this flow, we have a very small change in the cross-sectional area, we can then set $S' \approx b\zeta$, where b is the width of the channel at the surface. Applying this to (37) we arrive at

$$b \frac{\partial \zeta}{\partial t} + \frac{\partial(S_0 v_x)}{\partial x} = 0$$

By taking the first time derivative of both sides of this equation we obtain

$$b \frac{\partial^2 \zeta}{\partial t^2} + \frac{\partial}{\partial x} \frac{\partial(S_0 v_x)}{\partial t} = 0$$

which is possible due to the differential operator being commutative. Substituting the value for $\frac{\partial v_x}{\partial t}$ from (36) we obtain

$$b \frac{\partial^2 \zeta}{\partial t^2} - g \frac{\partial}{\partial x} \left(S_0 \frac{\partial \zeta}{\partial x} \right) = 0.$$

If the cross-sectional area of the fluid at equilibrium is the same throughout the channel we know that S_0 is a constant and therefore we can extract S_0 from the partial differential yielding

$$b \frac{\partial^2 \zeta}{\partial t^2} - g S_0 \frac{\partial^2 \zeta}{\partial x^2} = 0 \quad (38)$$

The equation we derived in (38) is called the wave equation. The wave equation provides us with a relation which we can use to determine the velocity of wave propagation, U . This relation is given by the square root of the coefficient of $\frac{\partial^2 \zeta}{\partial x^2}$ so for this wave, we have the velocity of the wave propagation is

$$U = \sqrt{\frac{g S_0}{b}} \quad (39)$$

We can see that the velocity of wave propagation for this system only depends on the width of the channel at the surface, b . It follows that for a narrower channel we will see that the waves propagated a larger velocity. Below is a plot to show the relationship between U and b .

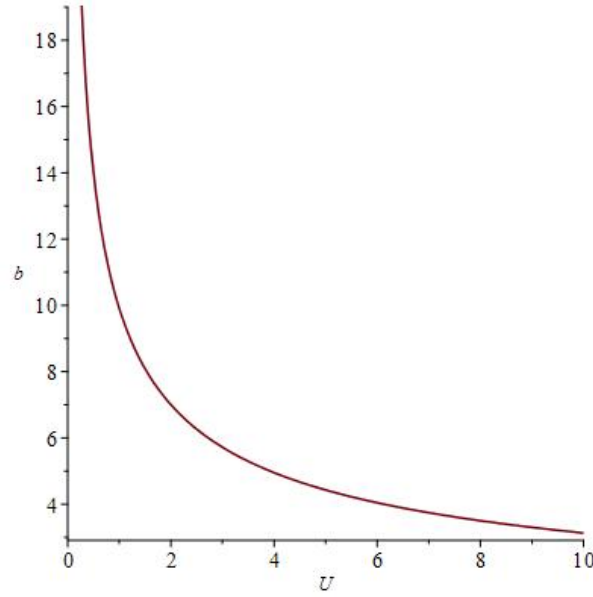


Figure 5 - Relation Between Velocity of Wave Propagation and Width of Channel at the Surface.

From this figure, we can see that the velocity of wave propagation decays rapidly for a small difference in b in narrow channels but for much wider channels a small change in b will yield a smaller difference in U . the reason in which we observe this property is due to the relation between pressure, p , and velocity, v . This relation can be observed in both Bernoulli's and Euler's equations. We can also justify the shape of the graph too, for much smaller widths of channels we will observe that there will be a larger pressure gradient and a small change in this width we will result in a larger percentage change in the value of the pressure hence our result of the figure for small values. A similar argument can be made for waves propagating in channels that are relatively large. This result can be seen in nature with the flow of water down a river, if you walk alongside a river you will observe that the velocity of the waves will be much larger in areas of the channel where it is narrower and much slower in areas where the river is wider.

3.4.3 Shallow Water Solution in a Tank

Using similar procedures as we did before in the 'Shallow Water Solution in a Channel' we can derive the solutions for the velocity of wave propagation in a large tank. The tank is assumed to be infinite in both the x and y -direction and the depth, the z -direction, has a depth which we shall denote by h . For this case, we know that the z component of the velocity, v_z , is small. This is similar to our previous case where both the y and z -components of velocity, v_y and v_z respectively, were both small in comparison to the x -component of velocity, v_x .

As the procedure to the beginning of this is so similar to the way in which we derived velocity of wave propagation in the previous subsection we shall not show the initial calculations for this exercise and begin by saying that Euler's equation for this problem takes the form

$$\frac{\partial v_x}{\partial t} + g \frac{\partial \zeta}{\partial x} = 0, \quad \frac{\partial v_y}{\partial t} + g \frac{\partial \zeta}{\partial y} = 0 \quad (40)$$

The equation of continuity is derived in a similar manner to before and it is given by

$$\frac{\partial h}{\partial t} + \frac{\partial h v_x}{\partial x} + \frac{\partial h v_y}{\partial y} = 0$$

Intuitively, we can express h as a sum of the depth of the fluid at equilibrium, h_0 , and the vertical displacement of the fluid from equilibrium, ζ . From these definitions, we obtain the following expression $h = h_0 + \zeta$. Substituting this into our equation of continuity we find

$$\frac{\partial \zeta}{\partial t} + \frac{\partial h_0 v_x}{\partial x} + \frac{\partial h_0 v_y}{\partial y} = 0. \quad (41)$$

If we now assume that the base of the tank is horizontal, $h_0 = \text{constant}$, and differentiate equation (27) with respect to t we get

$$\begin{aligned} \frac{\partial}{\partial t} \frac{\partial \zeta}{\partial t} + h_0 \frac{\partial}{\partial t} \frac{\partial v_x}{\partial x} + h_0 \frac{\partial}{\partial t} \frac{\partial v_y}{\partial y} &= 0 \\ \frac{\partial^2 \zeta}{\partial t^2} + h_0 \frac{\partial}{\partial x} \frac{\partial v_x}{\partial t} + h_0 \frac{\partial}{\partial y} \frac{\partial v_y}{\partial t} &= 0 \end{aligned} \quad (42)$$

Now if we substitute our equation (26) into (28) we get

$$\frac{\partial^2 \zeta}{\partial t^2} - g h_0 \left(\frac{\partial^2 \zeta}{\partial x^2} + \frac{\partial^2 \zeta}{\partial y^2} \right) = 0 \quad (43)$$

This is the wave equation and, as stated before, this equation corresponds to the velocity of wave propagation which in this case is given by

$$U = \sqrt{g h_0}$$

From this equation, we can conclude that the velocity of wave propagation in a tank will be a constant throughout the system as there is no free variable. The reason for this is due to there being no external factor which can impact the wave's velocity as the fluid is in a tank. Before when considering the velocity in a channel we found that the velocity of wave propagation only depended on the width of the channel. For this system, however, we note that the boundaries are all fixed in the tank so there will result in no change in the pressure hence our result.

3.5 Discussion of Results

This ends our analysis of gravity waves. Over the duration of this section, we have been able to evaluate the path of fluid particles and velocity of wave propagation at both the deep and shallow water limits. From the work that we have done in this section we have derived several different components which we will now take the opportunity to discuss the results that we found.

We begin with the path of fluid particles. In the analysis of these fluid motions, we derived the difference that the shallow and deep water limits have on the path of the fluid particles. We notice there is a notable difference between the two figures that we produced. We found that for the deep water limit the path of the fluid particle was given by a circle whose radius decreased exponentially for more negative values of z_0 . In the shallow water limit, we found the shape of the fluid particle to be an ellipse whose vertical component decreased and whose horizontal component remains the same as the value of z_0 became more negative. During the course of this analysis, we used two different methods to find the values for X and Z . We did, however, prove that the results we found were consistent in both cases. This is an important part of any scientific finding, the ability to reproduce finding using a different methodology. The reader will also be able to find consistency from other mathematicians work in this field, for example, the work done in [7].

The other major finding derived during this section was the velocity of wave propagation for three different systems. We found that for the first two systems we analysed, the function U dependent on different variables, and for the third case, U was found to be a constant. During the course of the section, we have provided a discussion of these results. Clearly from the results we would have liked to have provided the reader a derivation of the relation between velocity of wave propagation and the wave equation as this assumption provided a large jump in the derivation of the values for U , however, we felt that this derivation was outside of the scope of the project and if we were to continue with the project we would have considered adding this topic to the derivation. The reader will also be able to find a consistency of the results we found throughout the section in [5].

In the analysis of the gravity waves, in general, we have often created conditions to impose on the fluid to allow for the results to be simple. We did this to provide the reader with results which you would expect to find at this level of mathematics. If we would not have simplified the equations we worked with we would have found the project is more focused around the derivation as opposed to the results that we found. For a higher leveled mathematics paper clearly, we would have attempted to include both the deeper analysis and the discussion of the results.

4 Damping of Gravity Waves

Now that we have completed our section on gravity waves we will move the attention of the project towards a new section called damping of gravity waves. So far in this project, we have covered waves that have had very specific conditions imposed upon the fluid which has enabled us to be able to calculate wave propagation velocity and the equations of motion for these gravity waves. One of the conditions which we had imposed was that throughout the system the fluid had no viscosity. During this section, we will encounter fluids which possess larger values of viscosity in which we can not neglect the term.

In physics we will often encounter damping in many forms of different waves, in this

section we will clearly focus our attention on the viscous damping of fluid. Damping of fluid arises when there exists some internal friction between fluid particles. This internal friction results in the loss of kinetic energy in the system which yields the visual result of a wave's amplitude becoming smaller in time. By the end of this section, we hope the reader will have obtained an understanding of how we can calculate the value for the damping coefficient and how this coefficient impacts the shape of fluid motion in different fluids.

4.1 Oscillations in a Viscous Fluid

Before we can consider how damping effects gravity waves we must first develop an understanding of how oscillations in a viscous fluid behave.

Let us begin by again considering oscillations at the surface of a fluid, like the gravity waves we were studying in the previous section. We assume a body of fluid that is incompressible and bounded by an infinite plane surface which performs simple harmonic oscillations with frequency ω . We take the yz -plane to be the solid surface and the fluid region to be $x > 0$ and take the y -direction to be the direction of oscillations. The velocity of these oscillations in this instance will be given by u whose solutions will be given in the form

$$u = u_0 e^{-i\omega t}$$

where u_0 is a constant given by $Ae^{i\alpha}$. If we only consider linear operations upon the velocity, u , we can assume that the solutions will only take the real part so we can continue using the complex solution of the velocity but at the final solution we will consider only the real part, hence we have that

$$u_y = u = u_0 e^{-i\omega t}.$$

We also have that the fluid must satisfy the boundary condition at $x = 0$ given by $\mathbf{v} = \mathbf{u}$, hence at $x = 0$ we have that $u_x = u_z = 0, u_y = u$. It is clear that the oscillation is only dependant on x and time, t , so from the equation of continuity for incompressible fluids, $\nabla \cdot \mathbf{v} = 0$, we have $\frac{\partial v_x}{\partial x} = \text{constant} = 0$, from the boundary condition. As we have found that all terms are independent of y and z and v_x does not depend on x it follows that the advection term, $(\mathbf{v} \cdot \nabla)\mathbf{v}$, is also zero. From this the Navier-Stokes equation becomes

$$\frac{\partial v}{\partial t} = -\frac{1}{\rho} \nabla p + \nu \Delta v \quad (44)$$

which is a linear equation whose x -component is $\frac{\partial p}{\partial x} = 0$ or $p = \text{constant}$. It is clear that the y -component becomes $v_y = v$, this leaves us with

$$\frac{\partial v}{\partial t} = \nu \frac{\partial^2 v}{\partial x^2}. \quad (45)$$

This is the heat conduction equation. To obtain a solution of this we again consider the velocity to be in the form of

$$v = u_0 e^{i(kx - \omega t)}$$

and substituting this into the heat conduction equation (45) we obtain

$$-i\omega u_0 e^{i(kx-\omega t)} = -k^2 \nu u_0 e^{i(kx-\omega t)}$$

or

$$i\omega = k^2 \nu.$$

Solving for k gives us

$$k = \sqrt{\frac{i\omega}{\nu}} = \pm(1+i)\sqrt{\frac{\omega}{2\nu}}.$$

In this case, we only consider the positive solution due to the requirement of the velocity needing to decrease as you move into the fluid, so the solution of our velocity is

$$v = u_0 e^{-x/\delta} e^{i(x/\delta - \omega t)}$$

where $\delta = \sqrt{\frac{2\nu}{\omega}}$ which defines the depth of penetration for a wave in this system. We can see from this solution that transverse waves occur in viscous fluids however they are damped exponentially as we move away from the surface. We also see from δ with increasing frequency comes decreasing depth of penetration. The depth of penetration also increases with increasing kinematic viscosity.

4.2 The Damping Coefficient

In our previous section on gravity waves we had arrived at equations which represent the path of a fluid particle and velocity wave of propagation, however, we never considered the effect of damping on the wave propagation. During this section, we will work towards a solution to the damping coefficient of a gravity wave which allows us to have an extension of the solutions we had previously derived in the case where the fluid is viscous.

Let us now consider a more general case of oscillatory motion in a viscous fluid by this time taking a system which can take any shape. In our previous section we discussed motion that had an oscillating plane and during this section we found that the advection term $(\mathbf{v} \cdot \nabla)\mathbf{v}$ was identically zero this is not the case, however, for a surface which has an arbitrary shape so to begin this section we assume that the advection term is small in comparison to the other terms in the equation so that we can, again, neglect it from our derivations.

We begin by taking the curl of both sides of equation (44) which leaves us with

$$\frac{\partial \nabla \times \mathbf{v}}{\partial t} = \nu \Delta (\nabla \times \mathbf{v})$$

as the term $\nabla \cdot (\nabla p)$ is identically zero. Here we have that the vorticity of the fluid satisfies the heat conduction equation. We saw in the previous section that this results in an exponential decrease in the element which satisfies the equation hence the vorticity

of this fluid exponentially decreases as we move further into the fluid with the depth of penetration of the vorticity being of the order $\delta = \sqrt{\frac{2\eta}{\omega}}$.

We now arrive at a point where we have two limiting cases for this flow, these being $\delta \gg \lambda$ and $\delta \ll \lambda$, where λ is to the order of magnitude of the dimension of the oscillating body. We begin by considering the case where $\delta \ll \lambda$. As we have stated we are restricting ourselves to flow where the advection term is small in comparison to other terms so we must assume that the amplitude, a , is small in comparison to wavelength, λ , as we did before in the section on gravity waves.

Substituting our value of δ which we found in the previous section leaves us with two conditions

$$\lambda^2 \omega \gg \eta, \quad a \ll \lambda$$

where ω is the frequency. As the vorticity satisfies the heat conduction equation we can see the motion of the fluid is rotational in a thin surface layer of the fluid but through the rest of the fluid, we have potential flow.

We now wish to calculate the energy dissipation in a gravity wave. To do this we must consider the mechanical energy of the wave, E_{mech} which is given by the kinetic energy and the potential energy due to gravity. It is clear to see that $E_{mech} = E_{kin}$ in this case the presence of a gravity field will have no effect on the energy dissipation as we are considering energy dissipation which only results from the internal friction of the fluid. From this, we can use the following formula for $\dot{E}_{mech}$

$$\dot{E}_{mech} = -\frac{1}{2}\eta \int \left(\frac{\partial v_i}{\partial x_k} + \frac{\partial v_k}{\partial x_i} \right)^2 dV \quad (46)$$

It is important to note that because we are considering gravity waves we have a small region of the surface which has rotational flow and the velocity gradient in this region is large. Because of this property when we calculate the integral of a gravity wave we may ignore the surface of the fluid hence we are taking an integration over the whole volume of fluid which moves as if it was ideal. As we have a flow that is potential we know that $\mathbf{v} = \nabla\phi$, from this we can make a substitution for v_i and v_k in equation (46) to give us

$$\dot{E}_{mech} = -\frac{1}{2}\eta \int \left(\frac{\partial}{\partial x_k} \frac{\partial \phi}{\partial x_i} + \frac{\partial}{\partial x_i} \frac{\partial \phi}{\partial x_k} \right)^2 dV = -2\eta \int \left(\frac{\partial^2 \phi}{\partial x_i \partial x_k} \right)^2 dV \quad (47)$$

The potential, ϕ , we found has the form

$$\phi = \varphi_0 \cos(kx - \omega t + \alpha) e^{kz}$$

Substituting this value of ϕ into equation (47)

$$\dot{E}_{mech} = -2\eta \int \left(\frac{\partial^2 \varphi_0 \cos(kx - \omega t + \alpha) e^{kz}}{\partial x_i \partial x_k} \right)^2 dV = -2\eta \int (-2\varphi_0 k^2 \sin(kx - \omega t + \alpha) e^{kz})^2 dV$$

or

$$\dot{E}_{mech} = -8\eta k^4 \int (\varphi_0 \sin(kx - \omega t + \alpha) e^{kz})^2 dV$$

However we are not interested in the value of energy dissipation at one instant, we instead wish to find the mean value of the energy dissipation with respect to time. The mean value of energy dissipation is given by

$$\bar{\dot{E}}_{mech} = -8\eta k^4 \int \overline{(\varphi_0 \sin(kx - \omega t + \alpha) e^{kz})^2} dV$$

We know that the mean value of $\sin^2(x)$ is equal to the mean value of $\cos^2(x)$ so we can rewrite our equation for $\bar{\dot{E}}_{mech}$ as

$$\bar{\dot{E}}_{mech} = -2\eta k^4 \int \overline{(\varphi_0 \cos(kx - \omega t + \alpha) e^{kz})^2} dV$$

or

$$\bar{\dot{E}}_{mech} = -2\eta k^4 \int \bar{\varphi} dV \quad (48)$$

We can also find the value of $\bar{E}_{mech}$ from the theory that for any system that executes small oscillations the mean kinetic and potential energy are equivalent, using this and recalling that kinetic energy is given by $\frac{1}{2}v^2$ we can define $\bar{E}_{mech}$ to be twice the mean kinetic energy of the system which is given by

$$\bar{E}_{mech} = \rho \int \bar{v}^2 dV = \rho \int \overline{\left(\frac{\partial \varphi}{\partial x_i} \right)^2} dV.$$

If we substitute our value of the velocity potential into our previous equation we find

$$\bar{E}_{mech} = \rho \int \overline{(-\varphi_0 k \sin(kx - \omega t + \alpha) e^{kz})^2} dV.$$

By again using the definition that the mean values of $\sin^2(x)$ and $\cos^2(x)$ are equivalent we can simplify the equation to

$$\bar{E}_{mech} = 2\rho k^2 \int \bar{\varphi}^2 dV \quad (49)$$

To find the damping coefficient of these waves we now need to use the definition for the damping coefficient, γ , which is given by

$$\gamma = |\dot{\bar{E}}_{mech}| / 2\bar{E}_{mech}.$$

By substituting equations (48) and (49) into the definition of the damping coefficient we find

$$\gamma = \frac{8\eta k^4 \int \bar{\varphi} dV}{4\rho k^2 \int \varphi^2 dV} = \frac{2\eta}{\rho} k^2 = 2\nu k^2. \quad (50)$$

Finally by substituting the value of k with the value found in the derivation of the dispersion relation at the shallow water limit (19), we find that the damping coefficient for deep water gravity waves is given by

$$\gamma = 2\nu \frac{\omega^4}{g^2}. \quad (51)$$

For the shallow water limit we find that by substituting the dispersion relation at this limit into (50), the damping coefficient for shallow water gravity waves is given by

$$\gamma = 2\nu \frac{\omega^2}{gd}.$$

4.3 The Effect of the Damping Coefficient on Wave Motion

Now we have found the value of the damping coefficient all that remains is to see what effect this damping has on the waves that we have already discussed. To begin this we need only state the fact that over a period of time, t , the energy of a wave decreases according to the following law $\bar{E}_{mech} = C e^{-2\gamma t}$, where C is a constant. As is well known, the value of energy is proportional to the square of the amplitude hence the amplitude decreases at a rate of $e^{-\gamma t}$.

4.3.1 Deep Water Limit

Now we know what impact the damping coefficient has in terms of mathematics we will now find a visual representation of damped deep water gravity wave to gain a better understanding of this effect. As we know from the derivation of the damping coefficient there are three variables which can directly impact the value of γ for the deep water limit, however, we will here assume that $g = 9.81$ as usual. The two variables that we will manipulate are the kinematic viscosity, ν , and the frequency, ω .

To be able to visualise how the damping effects the wave motion we recall that the velocity potential for a gravity wave is given by

$$\phi = c_1 e^{kz} \cos(kx - \omega t).$$

By multiplying this value for the velocity potential by the rate of damping, which we found previously, we find

$$\phi = c_1 e^{-\gamma t} e^{kz} \cos(kx - \omega t)$$

or

$$\phi = c_1 e^{kz - (2\nu\omega^4 t)/g^2} \cos(kx - \omega t).$$

Which we can plot in maple, as usual. To plot this, however, we will consider water, $\nu = 0.01 \text{ cm}^2/\text{sec}$, and glycerine, $\nu = 6.8 \text{ cm}^2/\text{sec}$ so that we can assess a small and large kinematic viscosity for a variety of different frequencies, ω . We will here only consider waves which are at the surface, $z = 0$, and we will assume that $c_1 = 1$ and $k = 1$ for simplicity.

We first consider the fluid to be water and we will plot the following values for the frequency, $\omega = \{1, \pi, 2\pi\}$.

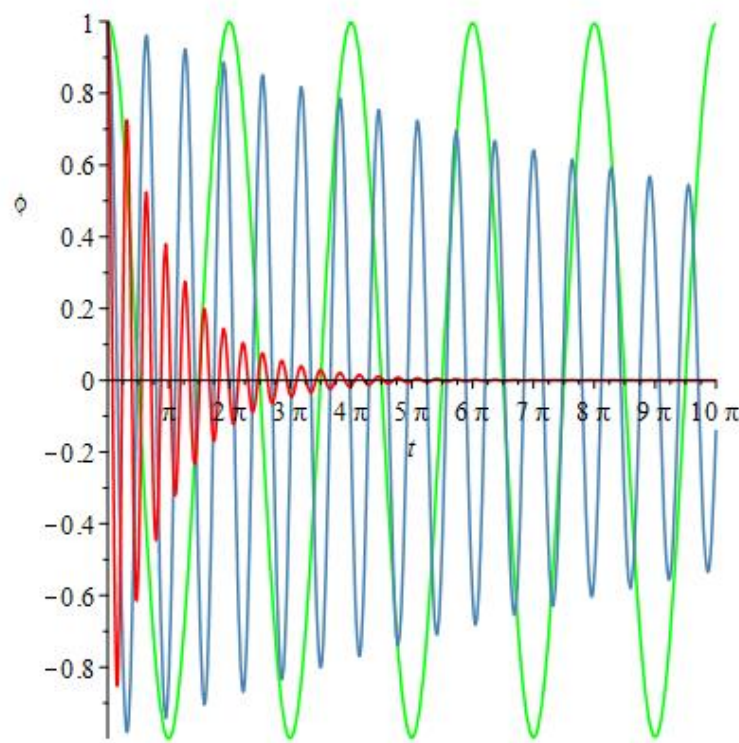


Figure 6 - Plot to Show the Velocity Potential as Time Progressed for a Deep Water Gravity Wave.

From this figure we can see that for fluids with a relatively small viscosity it requires a substantially large frequency to observe the effect of damping. We can see that for $\omega = 1$ you can not observe any damping in the fluid. We also note that once we start increasing the frequency in this wave, however, we begin to see the visual effect of damping. It is clear why this is the case by considering the formula for damping in these types of wave. Until the frequency has become large enough to compensate for the small value of $\frac{\nu}{g^2}$ the exponential will remain very close to 1 and hence we will see no change in the velocity potential. However, it is important to note that eventually over time waves of a lower frequency will be damped when time becomes large enough to

compensate for $\frac{\nu}{g^2}$.

We now consider the fluid to be glycerine and we will plot the following values for the frequency, $\omega = \{1, 1.2, 1.5\}$.

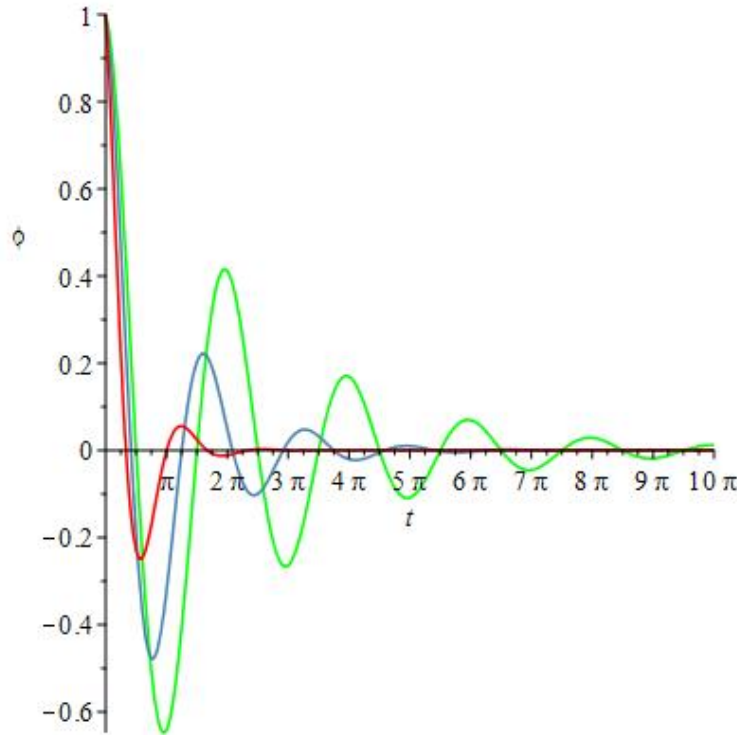


Figure 7 - Plot to Show the Velocity Potential as Time Progressed for a Deep Glycerine Gravity Wave.

Immediately from this figure, it is obvious that the damping has a substantial impact on the wave at lower frequencies of the wave. We see from the values we chose for ω that a small change in the value of the frequency results in an observable increase in the damping of the wave. This property is due to the frequency no longer needing to compensate for such a small value in the exponential.

4.3.2 Shallow Water Limit

Previously we found how the damping coefficient affected deep water gravity waves. In this subsection, we will see how the damping coefficient affects shallow water gravity waves. We begin by stating that the velocity potential for shallow water gravity waves is given by

$$\phi = \frac{gH}{2\omega} \sin(kx - \omega t).$$

The value for the rate of damping at this limit will be given by

$$e^{-(2\nu\omega^2)t/gd}.$$

Multiplying the velocity potential by this values will yield

$$\phi = \frac{gH}{2\omega} e^{-(2\nu\omega^2)t/gd} \sin(kx - \omega t).$$

This is a function which we can plot in maple. So that we are able to compare the effect between deep and shallow water waves we will use the exact same conditions that we used in the last subsection, $z = 0$, $k = 1$, $g = 9.81$. For this plot we have 2 extra constants which we define as follows $H = 0.1$ and $d = 1$. We will again consider two values for the viscosity, water ($\nu = 0.01 \text{ cm}^2/\text{sec}$) and glycerine ($= 6.8 \text{ cm}^2/\text{sec}$.) We first consider the fluid to be water and we will plot the following values for frequency, $\omega = \{1, 5, 10\}$

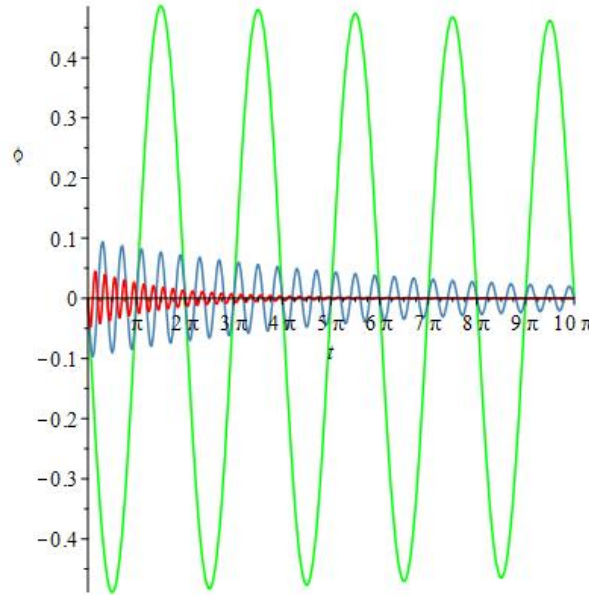


Figure 8 - Plot to Show the Velocity Potential as Time Progressed for a Shallow Water Gravity Wave.

From this figure, we can see that for shallow water waves we require a large value of the frequency for us to be able to observe a large amount of damping. We can see that when compared to the deep water limit the wave damping, in this case is, more linear and smooth. This is due to the value of the frequency of the deep water limit being to the power of four and the frequency at the shallow water limit being to the power of two.

Now we will consider the fluid to be glycerine and we will plot the following values for frequency, $\omega = \{0.3, 0.5, 0.8\}$

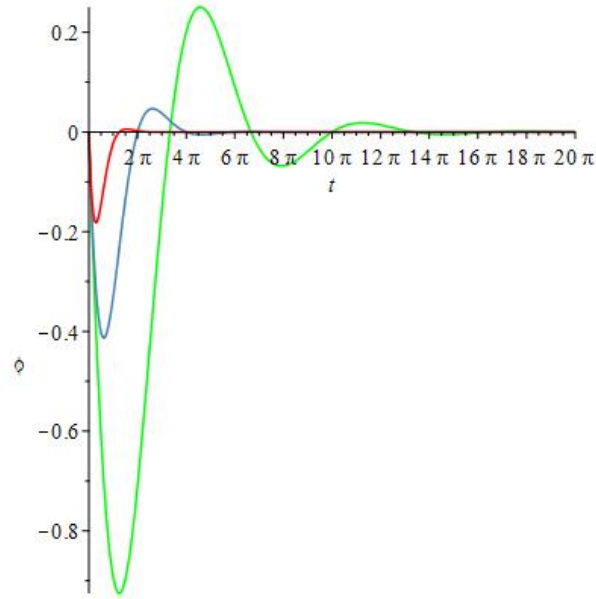


Figure 9 - Plot to Show the Velocity Potential as Time Progressed for a Shallow Glycerine Gravity Wave.

From the previous figure, we can observe the most critical form of damping that we have seen in this analysis. We see that for even small values of frequency the velocity potential of the wave is damped rapidly. This is due to the value of viscosity in this case completely cancelling the term gd hence the exponential will be very quickly become to the power of a substantial number as time progresses.

4.4 Discussion of Results

This ends our section on damping of gravity waves. During this section we have arrived at 4 figures to show how the dynamic viscosity, ν , and frequency, ω , impact the value of the damping coefficient and therefore the velocity potential of the waves in different systems. We will now take the time discuss the results we arrived at.

There are three major results that we have derived during the course of this section that comprise the overall findings: a value for the depth of penetration, The damping coefficient, the resulting damping of shallow and deep gravity waves for different values viscosity and frequency. We will begin our discussion on the depth of penetration. During this subsection, we found a result which signified how deep fluid movement can extend into the fluid below it. For this section, there is very little to discuss however you can see consistency with the results found in the section in [5]. The analysis of this equation required very little theoretical ideas it was simply a set of relations that we manipulated to find our result. This subsection did provide a very important finding at the end, however, which allowed us to continue and apply this definition to rest of the section.

Now if we consider the analysis we did in the subsection on the damping coefficient we again see that we required very little in terms of mathematical logic and we arrived at a value for two different coefficients. As we did in the derivation of the of the depth of penetration we simply manipulated equations which we previously knew to find our

equations of damping coefficient. The result of this subsection was vital for us to be able to continue in this section. You will find a consistency of these results across multiple different articles as this coefficient is a well know coefficient in physics.

Finally, in the last subsection of our analysis, we arrived at 4 different figures that corresponded to how different values of wavelength and viscosity affected the velocity potential of fluid at the deep and shallow water limit. During the research done for this project, we found that there was a very limited number of figures which showed this relation. In this analysis, we found that the for small values of frequency the damping of the velocity potential would substantially larger in the shallow limit than the deep limit for both values of viscosity. We also found that for higher values of frequency we would observe a larger value for damping in the deep limit than we did in the shallow limit. This effect, we found, was due to the power of ω in equations of the damping coefficient.

5 Capillary Waves

5.1 Introduction to Capillary Waves

Capillary waves, or ripples as they are more commonly known, are generated at the surface of fluid when there is a small disturbance to the surface of the water. Capillary waves consist of waves which have a very small wavelength, this property means that the capillary wave motion is determined mostly by surface tension at the surface of the fluid and has very little dependence on gravity, unlike the previous waves we have reviewed so far whose restoring force was gravity. As capillary waves are mostly influenced by surface tension they express rounded crests with a V-shaped trough. During this section, we will begin to derive mathematical formulas which can describe the path of a fluid particle in these capillary waves acting upon a plane.

5.2 Laplace's Formula

Before we begin to derive the equations of motion for a capillary wave we must first take in to account Laplace's formula as this formula allows us to calculate the surface pressure which will be an important component of a system. Surface tension, as previously mentioned, is very important for our analysis of capillary waves as their motion is determined mostly by surface tension. If we consider two media with differing pressure, air and water is a good example of this, and denote the pressure for the media as p_1 and p_2 respectively and let R_1 , R_2 be the principal radii of curvature at a given point of the surface, then the formula relating these components together is given by

$$p_1 - p_2 = \alpha \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \quad (52)$$

where α is the surface-tension coefficient. This is Laplace's formula. In this section, we will attempt to use this to find a formula to express the mechanical equilibrium of

two adjoining media. We assume that the media and surface separating the media have no external forces acting upon them. From this assumption we know that the pressures in both media are constant, hence equation (52) becomes,

$$\frac{1}{R_1} + \frac{1}{R_2} = C \quad (53)$$

Where C is a constant. This equation shows us that the sum of curvatures over a free surface of separation must be a constant and if the whole surface is free it must be spherical. We will now consider a surface that is supported along some curve, by a film of fluid, for example, then the result of this equation is not as simple. If we have the sum of $\frac{1}{R_1}$ and $\frac{1}{R_2}$ being the same everywhere we must also have the sums $\frac{1}{R'_1}$ and $\frac{1}{R'_2}$ where R'_1 and R'_2 are the radii of curvature on the opposite side of the film to R_1 and R_2 respectively. However, as R'_1 and R'_2 are on opposite sides of the film then they must have the same curvature but with opposite signs hence we have that.

$$\frac{1}{R_1} + \frac{1}{R_2} = C$$

and

$$-\frac{1}{R_1} - \frac{1}{R_2} = C$$

Which clearly can only be the case when $C = 0$, hence we have that,

$$\frac{1}{R_1} + \frac{1}{R_2} = 0$$

We now consider the case where we have a surface of a medium that is in the presence of a gravitational field. We will here consider the first medium to be an incompressible fluid and the second medium to be the atmosphere. We know that the first medium has no external pressure hence $p_1 = c - \rho g z$ where c is a constant. We also know that the second medium is a constant over the free surface hence $p_2 = \text{constant}$. Here z is measured, as usual, vertically upwards. We have for this system that the equilibrium condition is given by

$$\frac{1}{R_1} + \frac{1}{R_2} - \frac{\rho g z}{\alpha} = c \quad (54)$$

where c is again a constant and α is the surface-tension coefficient.

We now wish to determine the shapes of the surfaces which satisfy the equilibrium conditions found in (53) and (54). In order to be able to do this, we require formulae for the radii of curvature. We only desire to find the case where the surface has a small deviation from the plane, however, as a general case for this would require differential geometry. To begin we define the equation of the surface to be $z = \zeta(x, y)$, as we are considering small deviations we suppose that ζ is small anywhere on the surface. Using

the known formula for area, f , on a surface we have

$$f = \int \sqrt{\left[1 + \left(\frac{\partial \zeta}{\partial x}\right)^2 + \left(\frac{\partial \zeta}{\partial y}\right)^2\right]} dxdy.$$

However, as we assume that ζ is small everywhere we can approximate that the area of the surface can be given by

$$f = \int \left[1 + \frac{1}{2} \left(\frac{\partial \zeta}{\partial x}\right)^2 + \frac{1}{2} \left(\frac{\partial \zeta}{\partial y}\right)^2\right] dxdy. \quad (55)$$

We can also find the variation of the area, δf , which is given by

$$\delta f = \int \left[\frac{\partial \zeta}{\partial x} \frac{\partial \delta \zeta}{\partial x} + \frac{\partial \zeta}{\partial y} \frac{\partial \delta \zeta}{\partial y} \right] dxdy.$$

Integrating this formula by parts we find

$$\delta f = - \int \left(\frac{\partial^2 \zeta}{\partial x^2} + \frac{\partial^2 \zeta}{\partial y^2} \right) \delta \zeta dxdy. \quad (56)$$

We also have that the total change in area of separation is given by the formula

$$\delta f = \int \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \delta \zeta df \quad (57)$$

Comparing equations (56) and (57) we see arrive at

$$\frac{1}{R_1} + \frac{1}{R_2} = - \left(\frac{\partial^2 \zeta}{\partial x^2} + \frac{\partial^2 \zeta}{\partial y^2} \right). \quad (58)$$

This is the final form of the formula giving us a relation between the radii of curvature and the deviation of the surface from the plane for small values of ζ .

Our final part of the analysis on the Laplace equation will be to determine the boundary conditions that need to be satisfied for two fluids at the boundary. Of course, when we have two fluids both in motion at the boundary we will observe surface tension. For a theoretical system which would have no surface tension it is easy to see that at the boundary we would have the following condition $n_k(\sigma_{2,ik} - \sigma_{1,ik}) = 0$, which just simply states that the value of viscous friction in both fluids is equivalent to one another. If we now consider a system of two fluids which does possess surface tension at the boundary we must have that the difference between the viscous friction is proportional to the Laplace formula which will be acting along the normal, hence we have the

following boundary condition

$$n_k(\sigma_{2,ik} - \sigma_{1,ik}) = \alpha \left(\frac{1}{R_1} + \frac{1}{R_2} \right) n_i,$$

or

$$(p_1 - p_2)n_i = (\sigma'_{1,ik} - \sigma'_{2,ik})n_k + \alpha \left(\frac{1}{R_1} + \frac{1}{R_2} \right) n_i. \quad (59)$$

This equation gives us the boundary condition for two fluids at a boundary. Evidently, if both the fluids are ideal then, $\sigma'_{ik} = 0$ and we are left with the same equation as what we defined at the beginning of the chapter, equation (52).

We have just derived a boundary condition for this system of fluids, however, the equation that we found, (59) is not a general boundary condition. The reason for this condition not being general is due to the fact that surface-tension coefficient, α , is not necessarily constant over the surface of the boundary, this may be due to differing temperatures along the surface. To account for this deviation in surface-tension coefficient we are required to add a tangential force to the surface which is given by $\mathbf{f}_t = \nabla \alpha$ which is given per unit area of the surface of separation. For this system we will take the positive gradient as the force produced by the surface-tension will usually reduce the surface area. Applying this new force to equation (59) we find the boundary condition has become

$$\left[(\sigma_{2,ik} - \sigma_{1,ik}) - \alpha \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \right] n_i = (\sigma'_{1,ik} - \sigma'_{2,ik})n_k + \frac{\partial \alpha}{\partial x_i}.$$

An important property of this equation is that if we were to consider an ideal fluid, $\sigma'_{ik} = 0$, we would find that the left hand side of the equation would produce a vector along the normal and the right hand side would produce a tangential vector. Clearly, this is not possible unless both sides of the equation are equal to zero, so it follows that this equation is only true for viscous fluids.

5.3 Capillary Waves on a Surface

We begin this section as we did in the section on gravity waves by assuming that the amplitude of the waves is small in comparison to its wavelength and that the fluid satisfies the condition $\Delta \varphi = 0$. However, at the surface of the fluid, the conditions are now different to when we tackled gravity waves. We have that the pressure either side of the plane is different for this case, i.e $p_1 \neq p_2$. To begin we will use our findings from the section on Laplace's formula, more specifically (52) and (58), to find the conditions of the fluid at the surface. As before we use ζ to represent the displacement from equilibrium in the z -axis. Using (58) we can write Laplace's formula as

$$p - p_0 = -\alpha \left(\frac{\partial^2 \zeta}{\partial x^2} + \frac{\partial^2 \zeta}{\partial y^2} \right), \quad (60)$$

where p is the pressure near the surface and p_0 is the constant external pressure. We have already derived an equation for p in equation (12), by setting $f(t) = 0$ we have the following

$$p = -\rho g \zeta - \rho \frac{\partial \varphi}{\partial t}$$

Substituting this equation for p into (43) yields

$$\rho g \zeta + \rho \frac{\partial \varphi}{\partial t} - p_0 - \alpha \left(\frac{\partial^2 \zeta}{\partial x^2} + \frac{\partial^2 \zeta}{\partial y^2} \right) = 0.$$

As before we can omit the constant p_0 by redefining the potential, ϕ . Differentiating the previous equation with respect to t we find

$$\rho g \frac{\partial \zeta}{\partial t} + \rho \frac{\partial^2 \varphi}{\partial t^2} - \alpha \frac{\partial}{\partial t} \left(\frac{\partial^2 \zeta}{\partial x^2} + \frac{\partial^2 \zeta}{\partial y^2} \right) = 0,$$

and by making the substitution $\frac{\partial \zeta}{\partial t} = \frac{\partial \varphi}{\partial z}$, we obtain

$$\rho g \frac{\partial \varphi}{\partial z} + \rho \frac{\partial^2 \varphi}{\partial t^2} - \alpha \frac{\partial}{\partial z} \left(\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} \right) = 0$$

for $z = 0$. As before we make the assumption that the solutions to this equation are given by $\phi = Ae^{kz} \cos(kx - \omega t)$. Substituting this into the previous equation yields

$$\begin{aligned} & \rho g \frac{\partial Ae^{kz} \cos(kx - \omega t)}{\partial z} + \rho \frac{\partial^2 Ae^{kz} \cos(kx - \omega t)}{\partial t^2} - \\ & \alpha \frac{\partial}{\partial z} \left(\frac{\partial^2 Ae^{kz} \cos(kx - \omega t)}{\partial x^2} + \frac{\partial^2 Ae^{kz} \cos(kx - \omega t)}{\partial y^2} \right) = 0. \end{aligned}$$

Computing the values of the differentials and solving the result in terms of ω yields the following relation between ω and k given by

$$\omega^2 = \frac{\alpha k^3}{\rho} + gk. \quad (61)$$

We can conclude that when we consider long wavelengths such that $k \ll \sqrt{\frac{g\rho}{\alpha}}$, or $k \ll \frac{1}{a}$ where a is the capillarity constant, we can neglect the term representing capillarity in (61) and we are left with a pure gravity wave given by, $\omega^2 = gk$ which agrees with what previously found in the section on gravity waves. For short wavelengths, we can easily see that the effect of gravity can be neglected and hence we are left with the following relationship between ω and k

$$\omega^2 = \frac{\alpha k^3}{\rho} \quad (62)$$

Waves which are consistent with the previous result are called capillary waves or ripples, whilst waves with a wavelength that is neither long or short are called capillary-gravity waves.

5.3.1 Stoke's Drift Velocity

From this analysis one would assume that the motion of the fluid particle for a capillary wave would be similar to that of a deep water gravity wave, however, capillary waves in nature characteristically feature smooth round crests with a v-shaped trough. This is clearly counter-intuitive as the process of finding the motion of these particles would be identical to the deep water gravity wave and they both share a common value for the velocity potential, $\phi = Ae^{kz}\cos(kx - \omega t)$. The reason for this difference is due to a process called Stokes drift. Stokes drift describes the average velocity in which a material at the surface of a fluid will move during a time period. We will not derive Stokes drift velocity in this report, however, we will discuss the importance of this effect and how it impacts different classifications of waves. The horizontal component of Stokes drift velocity, $\bar{u}_s$, is given by

$$\bar{u}_s \approx \omega k a^2 e^{2kz}.$$

As we can see this formula is dependant on the value of the wave number, k , which itself is dependant on the wavelength, λ . It is evident from the Stokes drift formula that with an increasing value of the wavelength there yields an exponentially decaying value for the Stokes drift velocity. We must also consider the other variable that is featured in the formula, amplitude, as this will also affect the result of the Stokes drift velocity. Previously when we worked on deep water gravity we found that from our conditions the amplitude was found to be relatively small in comparison to the wavelength. During our discussion. we also found that the amplitude was small in comparison to the wavelength, so the value of a^2 cannot be the reason for the change in fluid motion. The amplitude is also an important variable when we consider the existence of the z component in the formula as this relates to the position of the fluid particle in the z axis. The reason we must consider this is that the component will differ largely in both deep water gravity waves and capillary waves as the amplitude can be considerably larger in deep water waves than it can be for capillary waves. If we consider both gravity waves and capillary waves with similar amplitude for simplicity, however, it is evident that capillary waves will be affected considerably more by this property than a deep water gravity wave due to the definition of a capillary wave stating that the wavelength is small in comparison to that of a deep water gravity wave. To be able to plot a relation between these values we first need to find the value for Stokes drift velocity in terms of the position along the x -axis. To do this we simply integrate the value of Stoke drift with respect to t , doing so yields

$$\bar{x}_s \approx \omega k a^2 e^{2kz} t.$$

A maple plot of this relation for an equal amplitude can be seen below. For this graph we have consider the following values for the wave number, $k = 10$ (to represent gravity waves) and $k = 20\pi$ (to represent capillary waves).

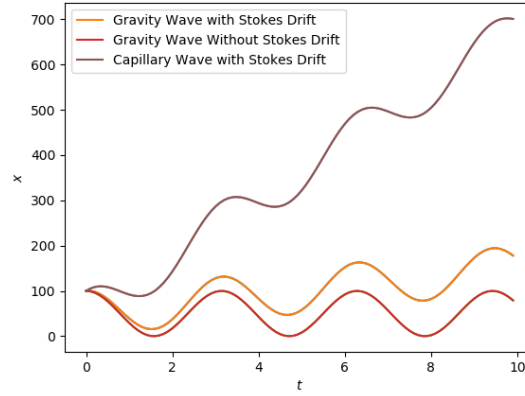


Figure 10 - A plot Showing the x Component of Three Different Waves as time Progress.

From this figure, we can see that the x component for capillary waves is considerably less affected by the Stoke Drift velocity than gravity waves. As we previously stated in the analysis, this is due to the value of wavelength being, by definition, much smaller for capillary waves which will yield a much larger value of the exponential factor than a gravity wave of the same amplitude. However, you can see that the Stokes drift velocity will also impact the horizontal component of gravity waves.

It follows then that the shape of capillary waves is due to the stoke drift velocity. When we consider this effect we observe that over a period of one oscillation the horizontal component has travelled a substantial distance. This distance accounts for the v shape you will see at the trough of the wave as near the trough the fluid particle's path is a very small circle. Near the crest of the waves, you will notice that the fluid particle's path is a very large circle. You will notice that this effect will also occur in gravity waves but to a much smaller extent and you will only notice a small difference between the position of the fluid at the start of the wave oscillation and at the end of the wave oscillation.

5.4 Damping of Capillary Waves

The only analysis remaining in our work on capillary waves is to calculate the damping coefficient for these waves. From this analysis, we will then be able to provide a discussion on how capillary waves are affected by damping compared to the other types of waves we have analysed throughout this project. To determine the damping coefficient of capillary waves we require only to make the substitution of our dispersion relation (62) into the formula for damping coefficient (50) which gives us the following

$$\gamma = 2\nu k^2 = 2\nu \left(\frac{\rho\omega^2}{\alpha} \right)^{\frac{2}{3}}, \quad (63)$$

this our required result for damping coefficient. As before when we consider damping we consider the amplitude of the waves to be damped at a rate of $e^{-\gamma t}$. To express this

visually we recall that the velocity potential for capillary waves is

$$\phi = c_1 e^{kz} \cos(kx - \omega t).$$

Multiplying the value of the potential by the damping term will result in

$$\phi = c_1 e^{kz} e^{-2\nu(\frac{\rho\omega^2}{\alpha})^{\frac{2}{3}}t} \cos(kx - \omega t).$$

As before we can plot this relation between frequency and velocity potential using Maple. We will again consider two values for dynamic viscosity, $\nu = \{0.01, 6.8\}$ for water and glycerine respectively. We will also consider the following values for the constants in the formula, $c_1 = 1$, $z = 0$, $k = 20\pi$, $\alpha = 1$, $\rho = 1$.

Our first plot we will consider the fluid to be water. By using the following values for frequency $\omega = \{1, 2, 5\}$ we obtain the following graph.

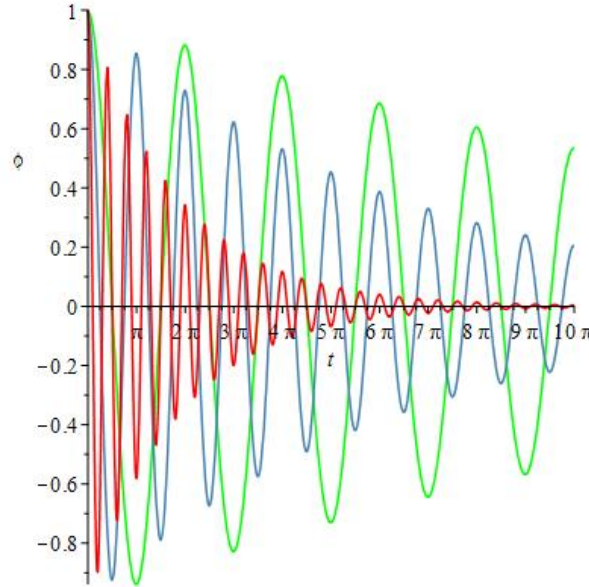


Figure 11 - Plot to Show the Velocity Potential as Time Progressed for a Capillary Wave in Water.

It is clear from this figure that we have a similar property for capillary waves as we did for the gravity wave before in water, that as frequency increases the damping of the velocity potential also increases. However, as when we compared the two types of gravity wave we found that shallow water gravity waves were more damped at lower frequencies and less damped at higher frequencies. This is the case, again with capillary waves. You can see at the initial small values of frequency the damping has a much larger effect than in gravity waves, however, with higher frequencies the rate of damping will be much less than a gravity wave. This is due to the term ω being technically to the power of $\frac{4}{3}$ in this damping coefficient which is much smaller than that of the gravity waves.

Our second plot we will consider the fluid to be glycerine. We will use the following values for frequency $\omega = \{0.05, 0.1, 1\}$

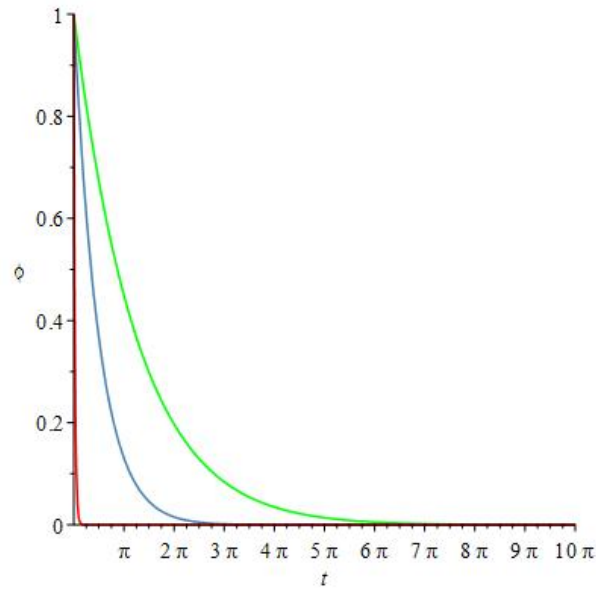


Figure 12 - Plot to Show the Velocity Potential as Time Progressed for a Capillary Wave in Glycerine.

Clearly, for this plot, we have a very strange shape of the damping. This shape takes credit from the large value of the viscosity which dominates the exponential. From these findings, you will see that a capillary wave will be damped rapidly and there will be no oscillations made by a capillary wave in a viscous fluid as the wave is damped quickly before it can penetrate into the fluid.

5.5 Discussion of Results

In this section, we began our analysis on capillary waves. Over the course of this section, we began by first finding different forms of the Laplace equation. This analysis provided us with the equations which determined the radii of curvature for a surface which displays surface tension. The analysis of the Laplace equation was very important for our further work with capillary waves.

Our main body of analysis in this section is the derivation of the equations of motion for a capillary wave. During this section, we derived the path of a fluid particle for this system. This section was very similar to the way in which we analysed deep water gravity waves. The important findings in this section was the work done on the Stokes drift velocity. This analysis allowed us to find a mathematical reason for the difference in the shape of the deep water gravity waves and capillary waves.

Our final analysis was on the damping of capillary waves. During this section, we found the coefficient of damping for capillary waves. From this analysis, we found that the damping behaved like the shallow water waves which we derived previously.

Across the literature, you will find that there are multiple resources that agree with the work done on the capillary waves motion and the damping coefficient. However, we will see that there are very few resources that discuss the effect of Stokes drift.

6 External Influences Acting upon the Fluid

Clearly, in reality, a system of fluid will not be as perfect in terms of conditions as we have considered them to be over the duration of this project unless the conditions have been manufactured to be so by humans. In this section, we will discuss the way in which these external factors, which we previously neglected, affect the motion of the fluid in a system. There are multiple factors which can affect the motion and of course we will be unable to discuss all of these factors in this project, however, we will discuss how an adsorbed film on the surface of the fluid and how a wind travelling over a wave will affect a previous work on waves.

6.1 The Effect of an Adsorbed Film on the Surface

We will now consider a surface of a fluid which has a film of adsorbed material on top of it. Immediately, we can see that the presence of this film will result in a change of the hydrodynamic properties of the fluid. This change is a result of the film being stretched or compressed as the motion of the fluid changes the shape of the surface hence the concentration of the adsorbed material will differ along the surface of the fluid. The result of this difference in concentration correlates to additional forces which must be regarded when considering the boundary conditions of the system.

For simplicity in this section, we will only consider adsorbed films that are insoluble in the fluid, for example, oil and water. We choose the film to be insoluble as if we chose a soluble film we would then have to consider diffusion of the film into the body of fluid. This is especially difficult to do analytically when we recall that the concentration of the film will differ over time and over the surface.

Previously when considering fluids which have surface tension, like we did in capillary waves, we considered the surface-tension coefficient to be a constant. For a fluid with an adsorbed film on the surface, it is not the case. The reason it is not a constant for this system is due to the fact that α is a function of surface concentration. It follows that if this surface concentration, γ , changes along the surface then α is also a function of the surface coordinates. From this, we can see that for our system we will have a tangential force which will need to be accounted for in the boundary condition. We previously discussed tangential forces for systems of two fluids in our section on Laplace's equation, here we found that the tangential force was given by $\mathbf{f}_t = \nabla\alpha$. However, for our system, we know that α is a function of the surface concentration so it follows that we can express $\nabla\alpha$ in terms of γ by some manipulation of partial differentials shown below

$$\mathbf{f}_t = \frac{\partial\gamma}{\partial\gamma} \left(\frac{\partial\alpha}{\partial x_i} \right) = \frac{\partial\alpha}{\partial\gamma} \left(\frac{\partial\gamma}{\partial x_i} \right).$$

As we discussed before this equation is only satisfied when the fluid is viscous. From this condition, we can see that for fluids that have a very low viscosity we can disregard the presence of the film and consider the fluid to be moving as normal.

We now begin our analysis to find the motion of a fluid which has been covered by

a film. So far we have found the equations of motion and have determined the boundary condition which must be imposed for this system. We now desire one more formula as we have one more unknown variable, γ , this final equation is an equation of continuity, an equation that we have previously found two versions of to work in separate cases. The equation of continuity for this system must express the total amount of the adsorbed material is unchanged in time. We must also consider the shape of the surface when finding the equation of continuity for this system too as this will impact what form the equation takes. For simplicity, we will consider the shape of the surface to be a plane which yields the following equation of continuity

$$\frac{\partial \gamma}{\partial t} + \frac{\partial \gamma v_x}{\partial x} + \frac{\partial \gamma v_y}{\partial y} = 0,$$

where, in this case, all the values are considered to be at the surface.

One interesting effect that this adsorbed film will have is given by the motion of a gas bubble in a viscous fluid. If the bubble has no film on the surface then the gas inside the bubble is free to move which results in there being less drag exerted on the bubble by the liquid as if it was a solid object of the same shape. If the bubble has a film of adsorbed material coating the surface, however, we will notice that the gas inside the bubble will remain at rest, hence a bubble completely surrounded by a film on the surface will move through the fluid as if it were a solid.

We will now attempt to find how this film will affect the damping coefficient for capillary waves. To do this we will consider a fluid whose viscosity is relatively small and hence the fluid can be assumed to be incompressible. We can make this assumption because for a fluid with a low viscosity the tangential forces acting on the fluid will be small. From these assumptions, we can calculate the energy dissipation of the fluid as if we were considering a fluid with a solid wall at a boundary. To begin our analysis we define the velocity to be of the form

$$\phi = \phi_0 e^{i(kx - \omega t)} e^{-kz}.$$

The energy dissipation per unit area of the surface is known to be given by

$$\bar{E}_{kin} = -\sqrt{\left(\frac{1}{8}\rho\eta\omega\right)|k\phi_0|^2}.$$

The total energy is given by

$$\bar{E} = \rho \int \bar{v}^2 dz = \frac{1}{2k} \rho |k\phi|^2.$$

The damping coefficient is therefore given by

$$\gamma_1 = \frac{\omega^{7/6} \nu^{1/2}}{2\sqrt{2\alpha^{1/3} \rho^{7/6}}} = \frac{k^{7/4} \nu^{1/2} \alpha^{1/4}}{2\sqrt{2\rho^{7/4}}}.$$

Comparing this with our damping coefficient for a capillary wave with a clean surface (63) we can see that the effect of damping is larger in fluids which have layers of film of the surface for most fluids flows, except the case when the wavelength is considerably small.

6.2 The Effect of Wind Passing over the Surface

Previously when we have considered waves we have considered how each one that we have defined (capillary, deep water and shallow water) as separate entities. However, in nature, you will tend not to find the wave separated from one another. The reason for the production of most waves we find in nature is due to external factor wind, which we have not considered at all during this project. If we now consider the process of wind we can begin to develop an understanding of how waves are generated and how prolonged periods of time can cause a change in the definition of the wave.

6.2.1 Pressure Effect

There are multiple properties of wind passing over the surface of a fluid which results in the production of capillary waves. The factor that will discuss in this section we shall denote as the pressure effect. The pressure effect arises when a change in the wind speed passing over a surface fluctuates. This fluctuation in wind speed will result in the air pressure above the surface changing over time. During our analysis of capillary waves, we found a formula to relate the difference in pressure in two media to the radii of curvature this formula is given by

$$p - p_0 = -\alpha \left(\frac{\partial^2 \zeta}{\partial x^2} + \frac{\partial^2 \zeta}{\partial y^2} \right),$$

where here p_0 is given by the constant internal pressure and p is given by the pressure at the surface. This was the first equation that began our analysis of capillary waves so it follows that this change in pressure due to the change in wind speed results in a production of capillary waves at the surface. We can extend this further by expressing a value for the pressure at the surface. To do this we must consider a form of Bernoulli's equation,

$$p = c - \frac{\rho}{2} v^2 - g\rho z.$$

From this, it is easy to observe that the pressure will decrease at velocity increases. From this observation, it follows that for flows of wind that have a relatively high velocity we will find a greater difference between the pressure in the atmosphere and the internal pressure. This implies that when the wind blows over the surface the radii of curvature is dependant on the velocity of the wind. More specifically the larger the ve-

locity of the wave the larger the radii of curvature for the surface.

Another important feature of these wind-generated waves is that the shape of the capillary waves amplifies the pressure difference between the two adjoining media. This amplification is due to the velocity of the wind increasing at the surface as the wind passes over the crest of a capillary wave, hence creating a lowering the pressure further at this point. Similarly, the velocity of the wind passing over the surface at the trough will decrease hence resulting in a larger pressure at this point. The reason for this change in velocity is due to the fact that as the wind approaches a crest of a wave the wind will find a blockage in its natural horizontal flow forcing the air to travel over the wave, at the top of the wave there will be horizontal flow along the surface of the same velocity which will combine with the air attempting to pass over the wave which results in a higher overall velocity at the crest. A similar argument can be made about the reduction of the wave speed at the trough due to the more space the wind can move into at the trough.

6.2.2 Growth of Waves

In nature, we find that capillary waves and gravity waves are not generally produced separately. Usually, gravity waves are generated by prolonged periods of time in which the wind passes over the surface of capillary waves. There are multiple theories which attempt to explain this feature, however, we will consider theories during this project.

In general, there are three major components which influence the growth of gravity waves. These three components are given by the wind speed, the duration in which the wind is flowing over the surface and the fetch, which is defined as the distance that the wind flows over. With these factors in mind, we will continue to look at the theories of wave growth.

We begin our analysis with the theory posed by Australian geophysicist Owen Martin Phillips in [12]. In this article, Phillips discusses his theory on the generation of waves by turbulent wind. The main findings in this article are that turbulent eddies that are found in the wind will exert random stress fluctuations on the surface. From the eddies found in wind, you will observe a fluctuation in the pressure at the surface, these pressure fluctuations will themselves vary in magnitude and frequencies. From these pressure fluctuations, it is said that the wave will begin to develop and grow into the gravity waves that we have discussed previously. This is very similar to what we considered earlier in terms of the pressure difference producing capillary waves. It seems then that with the help of some turbulent wind the capillary will continue to gain kinetic energy until the wavelength has become sufficiently large to where we can consider it a gravity wave.

The other theory that will consider is the theory posed by the American professor for applied mechanics John W. Miles in [11]. In this article, Miles posed his theory in the way in which wind accounts for the growth of waves. Miles theorised that as wind flows over a moving wave there was a development of complex air system above the surface of the wave. This complex system of air implies that for wind speed that is smaller than the celerity of the wave the wind would flow in the opposite direction to which the wave

is propagating. If the wind speed is larger than the wave celerity the wind will flow in the direction of the wave. These two factors would result in a circulation of motion around the surface of the waves. This circulation would then provide the waves with an increase in the amplitude of the wave it was theorised. From what we discussed earlier about the increase and decrease of velocity at the crest and trough signifies that these circulations would occur between above a trough of a wave.

7 Conclusion

This brings our main analysis of work to an end. Throughout this project, we have begun developed an understanding of the characteristics of the fluids we observe on a daily basis. This project has attempted to begin an in-depth analysis of the equations that mathematically determine the flow of different types of waves in varying systems of fluid. The main findings of this project were the derivation of the paths of fluid particles and the velocity of wave propagation in both shallow and deep system of fluid. In doing so we observe the effect that these different limit had on these two characteristic properties of the waves. Given the opportunity to continue with this project we would have continued with this work to consider the limit where the wave is considered to be in a system that is neither considered to be a shallow or deep limit. This derivation would have allowed the reader to have a full understanding of every possible system which can be seen in nature.

Another major result of this project is the damping effect of gravity waves. During this analysis, we were able to derive an equation which defined how the gravity waves were damped. In this analysis, we again observe the difference between shallow and deep limits for different values frequency and viscosity. If we were to continue this project we will have discussed the damping of these gravity waves for the internal flow of the fluid, doing this will have provided a better understanding of how all of the fluid in a system behaves under the effect of damping.

The main discussion of work in this project was the work done on capillary waves. During the discussion of capillary waves, we were able to find the path of a fluid particle in these waves and compare them to that of a gravity wave. In doing so we came to a conclusion that with no external factors this motion would be the same. However, when considering the Stokes drift velocity we were able to distinguish why the capillary waves naturally have a different looking shape to the gravity waves. To continue with this section, we would have liked to have been able to derive this stokes drift velocity as it would have provided a better understanding of the effect in general. In this section we also found the value for damping coefficient in capillary waves. This was an important result as it enabled us to compare this coefficient with the ones derived previously for gravity waves and again note the difference between the two types of waves. This section enabled to characteristically define the differences between capillary and gravity waves.

Our final section of this project took a more discussion like approach than our previous analysis of topics previously. The reason for this is due to the complexity of the

topics that we were discussing in this section. Clearly, if we were to continue we would like to take the opportunity to really discuss the processes considered in this section with mathematical derivation. The first major result of this topic was the discussion of how a film on the surface would impact the characteristics of the capillary waves which we had found before. The other main result of this section was the effect that wind flowing over the surface contributes to the generation and growth of waves. Clearly in this section the area we would focus most of our attention to is the growth of the waves.

This brings an end to this project we hope that you have enjoyed reading.

8 Acknowledgements

I would like to take this opportunity to give thanks to my supervisor [REDACTED] [REDACTED] for his involvement in this project as well as all of the staff at the University of Lincoln School of Mathematics and Physics, with special thanks to [REDACTED] [REDACTED] for the support that they have provided me to complete this project.

I would also like to give a huge thanks to my friends and family for all the support they have given over the course of the year, specifically my mother, my grandmother, my father, [REDACTED].

I would like to give my thanks to Everyone at the University of Lincoln Library for being so accommodating whilst writing this project, with an extra special thanks to Imran Muhammad for being a fantastic supervisor at work.

Finally, I would like to make a very special thanks to both [REDACTED] [REDACTED] for providing me with the encouragement to complete this project to the highest possible standard I could achieve and always being interested in what I was writing about even if they didn't understand it.

9 Appendices

9.1 Code

9.1.1 Code for Plot of Stokes Drift

```
#include "stdafx.h"
#include "matplotlibcpp.h"
#include <vector>
#include <map>
#include <numeric>
#include <algorithm>
#include <stdexcept>
#include <iostream>
#include <stdint.h>
#include <math.h>
namespace plt = matplotlibcpp;

int main()
{
    int Npoints = 100;
    double w = 1, k = 10, a = 1, z = 0, e = 2.718;
    double k2 = 2 * 3.14 / 0.1;
    std::vector<double> x(Npoints), y1(Npoints), y2(Npoints), y3(Npoints),
    y4(Npoints), y5(Npoints);

    for (int i = 0; i < Npoints; i++) {
        x[i] = double(i) / Npoints * 10;
        y2[i] = pow(-(k / w)*cos(-w*x[i]), 2) + w*k*a*a*x[i];
        y3[i] = pow(-(k / w)*cos(-w*x[i]), 2);
        y4[i] = pow(-(k / w)*cos(-w*x[i]), 2) + w*k2*a*a*x[i];
    }
    plt::plot(x, y2);
    plt::named_plot("Gravity Wave with Stokes Drift", x, y2);
    plt::plot(x, y3);
    plt::named_plot("Gravity Wave Without Stokes Drift", x, y3);
    plt::plot(x, y4);
    plt::named_plot("Capillary Wave with Stokes Drift", x, y4);
    plt::xlabel("$t$");
    plt::ylabel("$x$");
    plt::legend();
    plt::show();

    return 0;
}
```

9.2 Research Plan

9.2.1 Brief Description of the Project

This project will discuss the Mathematics that are related to the natural process of rippling water. I will discuss what causes waves to appear in oceans and how we can use equations to show the properties of the waves and predict their path. Towards the end of the project I will also discuss how external factors can affect these waves, for examples what wind does to gravitational waves or how an oil spill can affect how waves are formed.

9.2.2 Reasons for Choosing the Project

I chose this project several reasons one of the being the fact that whilst researching the project and beginning to write my findings down I will also be studying the fluid dynamics course which goes hand in hand with the project I have chosen and will help vastly not only in the fluid dynamics project but also in the mathematics project at the same time. This was not the main factor in choosing this project however what really drove me to choose this project was the fact that many great mathematicians for their dissertations also studied the topic of fluid dynamics and the module as a whole is regarded with respect in the community so, of course, I would like to produce a third year project that shows employers and fellow mathematicians that I am competent at mathematics and can write a piece on a difficult area of the subject.

9.2.3 Connections with Previous Modules

The mathematics in this project is very strongly related to the third year course Fluid Dynamics and most of the work I will be doing in the first three months will be applying what I have learnt in those lectures to this project. This project is also tied to Calculus, Differential Equations and Complex Analysis but this is less to do with the content in these modules and more down to the mathematical methods learnt in these modules that will help me to understand and manipulate the equations found in this module. You may also consider Methods of Mathematical Physics to be tied to this project as Green's theorem for example will appear in both instances and the module will also help me manipulate more complex equations and functions even if the ones being analysed in the Methods of Mathematical Physics don't directly apply to my project.

9.2.4 Survey of the Literature

The three books I will be using mainly is elementary fluid dynamics, an introduction to fluid dynamics and fluid mechanics. These books will be used for the bulk of the project and the equations used throughout the report will be taken from these books. There may need to be some more specialised books towards the end of the report but to begin there is more than enough material in here. The other book in my references, computational methods for fluid dynamics, will be used for any computational mathematics that will be used in the report.

9.2.5 Action Plan

This project will coincide with the fluid dynamics module and so it difficult to give estimates as to what will be completed at what time as what I learn in that module will almost always be important information on my project, however I will give some brief deadlines that I would like to stick to at the very least, listed below:

October – By the end of this month I will have the Latex document properly formatted with the specifications of the module handbook so that after this all I have to do is write the project inside the template. I also hope achieve a significant amount of research and to begin adding my references and appendices to my project ready for writing the report.

November – By the end of this month I will have a section added in my report following what was learnt in the lectures and what research I found relating to that found in the previous month, this section will probably be on the equation of continuity, Euler's equation and Bernoulli's equation. This will be necessary as the equations underpin this report so full derivation and manipulation of the equations is what I hope to have included in this report by the end of the month.

December – This month I'm less clear on what I want to have achieved by the end of but I would like to try and a lot more bulk into the report at this point so just writing up whatever I have found from research and lectures will be the bare minimum I hope to achieve in this month. Possibly adding my introduction as by this point I will almost know everything I need to know for this report.

January – At this point I will need to have my progress report wrote up so this is the aim for the end of this month.

After this point, there will be a progress report where I can re-evaluate where I am at in the report and can set some more deadlines for the report. I have discussed with [REDACTED] and we have agreed to have weekly 25 minute meetings to discuss the project.

9.3 Progress Report

9.3.1 Introduction

This document has been made to be given as an indication of the progress I have made during the first semester of my report. Through this document, I hope to inform the reader of what I have achieved thus far in the project showing and explain the sections that have been completed with the most important parts of the theory being implemented into this document. I also hope to show how well I have stuck to the research plan originally made at the start of the module and what alterations have been made to the plan since to accommodate the new information I have on the topic. The research plan that I had made at the start of the module has been stuck to very well as I have managed to stick to all the deadlines I made for myself each month and also being able to add more work on top of this. In hind sight, the targets I had set myself at the start of the module were clearly not difficult to stick to and I could have set myself bigger targets due to how quickly the content in fluid dynamics was taught. This being said I still believe my research plan was good for what it did at the time as if I had stuck to the original plan I would have a solid basis of which to work from. The extra work I did do for this project was mainly proposed by [REDACTED] our meeting and because of this, it would have been very difficult to add this to a plan which I had made after only a few meetings.

Analysis of work

Thus far in the module, I have completed a section on the prerequisite understanding that will be needed from the module fluid dynamics for someone reading the report will need to possess in order to understand truly the mathematics that has been added

further into the report. I believe I have completed this section but work could maybe still be done on it. So far in this section, there is a subsection which goes through the derivation of the equation of continuity, Euler's equation, Bernoulli's equation, potential flow, tensor form and Navier Stokes equation. Adding this section not only helped to explain to the reader what they needed to understand it also gave me an easy way in which to reference the equations like Navier Stokes later in the report where I would be using them as a opposed to defining the equation and then applying it.

The other section of the report that I have completed to this point is the section on gravity waves. This section is the very important in this project as it shows the reader how to be able to mathematically represent a wave in the simplest of cases, by doing this it allows me to then bring in more complicated ideas and conditions to a wave and then attempt to solve these, for example, viscous fluid. The contents of this section have been written closely alongside my own learning for the Landau book. The subsections in this section contain how you would use linearisation to simplify the equation of a wave and how we can manipulate this equation to give us a direct solution for the flow potential and hence the velocity of the wave in small velocity waves and long gravity waves.

Finally, the section of the report I am working on now is the section on damping of gravity waves and thus far for this section, I have read multiple resources as so to gain an insight on to the section and have started to write up the section. So far in this section, I have arrived at the equation for the derivative of the mechanical energy in the system and am currently looking to arrive at the damping coefficient by the end of the month as I have already read on the subject and understand it.

9.3.2 Gravity Waves

Findings

As discussed during the introduction the first main section of my report so far is on gravity waves. This section of the progress report will take a look at what I have done towards this topic in my report currently and what alterations still need to be made to it to be perfected. What I have done so far for this section of the module is by beginning with Euler's equation of a fluid which is defined as,

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = \nabla p + \mathbf{g}$$

then from this we use linearisation so that we can neglect the term $(\mathbf{v} \cdot \nabla) \mathbf{v}$ from the equation as we find it to be much less than the term $\frac{\partial \mathbf{v}}{\partial t}$. From this, you can then redefine the potential to be

$$\varphi - \left(\frac{p_0}{\rho} - C \right) t$$

which can be done without changing the results as the other term is a constant with respect to x, y and z . Doing this and defining the following

$$v_z = \frac{\partial \zeta}{\partial t} = \left(\frac{\partial \varphi}{\partial z} \right)_{z=\zeta}$$

to be true by definition we can set the two results equal to one another as they are both equal to zero. Doing this and using that $\Delta \varphi = 0$ from definition of a flow potential we have a pair of simultaneous equations,

$$\left(\frac{1}{g} \frac{\partial^2 \varphi}{\partial t^2} + \frac{\partial \varphi}{\partial z} \right)_{z=0} = 0$$

and

$$\Delta \varphi = 0$$

which we can use to find the solution for the potential. If we assume that the solution for the wave is given by,

$$\varphi = f(z) \cos(kx - \omega t)$$

then by using the simultaneous equations we found earlier we can solve for potential to give,

$$v_x = -A k e^{kz} \sin(kx - \omega t), \quad v_z = -A k e^{kz} \cos(kx - \omega t)$$

and also we have the solutions

$$x - x_0 = -A \frac{k}{\omega} e^{kz_0} \cos(kx_0 - \omega t)$$

$$z - z_0 = -A \frac{k}{\omega} e^{kz_0} \sin(kx_0 - \omega t)$$

These are results for gravity waves whose amplitude is very small in relation to its wavelength, I have also done a section on long gravity waves as well which is defined as the length of the wave being large in comparison to the depth of the fluid. I follow similar procedures during this section which eventually arrives to,

$$\frac{\partial^2 \zeta}{\partial t^2} - g h_0 \left(\frac{\partial^2 \zeta}{\partial x^2 - \frac{\partial^2 \zeta}{\partial y^2}} \right) = 0$$

Conclusions

I feel this section in terms of mathematical content is complete, the only things that I need to work on now is to go through the section and include graphs of the solutions using maple. Other than plotting the solutions I feel as if the section is as good as done, however, there is a possibility I will need to use C++ somewhere in here to solve a differential equation but this will be discussed with [REDACTED] once I have been taught the appropriate material.

9.3.3 Damping of Gravity Waves

For this section, i don't feel I have yet written enough on the topic to be able to do a write up like the previous section right yet but I will discuss what I still need to do in this section. So far in this section of the report, I have only been able to get up to writing an equation for the time derivative of mechanical energy which is given as,

$$\dot{E}_{mech} = \frac{1}{2}\eta \int \left(\frac{\partial v_i}{\partial x_k} + \frac{\partial v_k}{\partial x_i} \right)^2 dv$$

This is as far as I have got in this section. The things that still need to be added are just the derivation of the damping coefficient which is defined as,

$$\gamma = \frac{2v\omega^4}{g^2}$$

and also need to go through and number the equations I have already began writing. I should have no issue in completing this before Christmas and then I feel I will be on track to have the report done to a high standard in time for the deadline.

9.3.4 Action Plan

In this section of the midway review, I am going to write up the plan I have for the rest of the year, as said in the first research plan that was submitted. This plan will give me the necessary guidelines to attempt to stick to for the remainder of the university year to allow for completion of the project to its highest possible standard.

January

This month I hope to have the section on damping of gravity waves to be completed which will hopefully be done at the beginning of the month to allow for other work to be done. This month I also hope to be able to obtain quality graphics for my first two sections using maple, ANSYS and possibly C++. Doing this would not only give the reader a visual representation of the solutions I have found but also allow me to easily do this again for other solutions for similar graphs that may be required in future sections of the project.

February

This month I aim to discuss with [REDACTED] what my next section should be on. Once we have discussed this new section I would realistically complete the research needed for the topic by at least halfway through the month and begin adding this section to my report.

March

This month I want to be able to finished the section previously started in February

to a high standard and then again discuss with [REDACTED] what I should be moving on to next with this project, at this point I am hoping to be near the end of the basic topics of discussion and to be able to move on to more complex areas with preference to doing research into more applications of the solutions I have found and seeing how external factors can affect the solutions I have been producing.

By this time I expect my numerical methods lectures will have advanced to a point in which we have been taught how to use the Runge-Kutta method of solving differential equations, this being the preference of [REDACTED] some differential equation in the report.

April

This month I hope to be coming near to the end of researching new topics to add to my project and having a solid idea of what topics I will be adding to the project in the near future. This month I also wish to be able to start writing up most of these sections.

May

Early this month I hope to be finished with my report to give time for it to be proofread by another person and to gain feedback on my report. Once I have the results back from the proofreader I aim to be finished completely ready for submission after any alterations need to be made.

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